
5G Microstrip Patch Array Design and Beam Steering

Analytical Design and CST Full-Wave Simulation

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Independent Engineering Portfolio Project

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Portfolio Context and Evidence

Project Context. This work originated from a graduate-level antenna engineering project completed during Summer 2026. The present document is an independently prepared public portfolio edition intended to demonstrate analytical antenna design, full-wave electromagnetic simulation, array analysis, beam steering, engineering interpretation, and technical documentation.

Public Portfolio Edition. The original academic report has been reorganized and edited for professional and educational use. Course-specific grading material, student-identification information, institutional branding, and administrative content have been removed. This document does not represent an official publication or endorsement by any university, instructor, software vendor, or other institution.

Evidence Classification. The results presented in this report are based on analytical calculations and CST Studio Suite full-wave simulations. No antenna prototype was fabricated, and no experimental measurement campaign was conducted. Accordingly, the reported resonant frequencies, scattering parameters, radiation characteristics, efficiencies, directivities, and beam-steering results are classified as analytical or simulated evidence, not measured performance.

Simulation Scope. The baseline models were developed using CST Studio Suite Learning Edition. Because of the available mesh-cell limitation, PEC metallization was used for the successfully tuned baseline simulations while dielectric substrate loss was retained. A supplementary fixed-mesh finite-conductivity copper simulation was performed for the single patch to evaluate conductor-loss sensitivity. The numerical limitations and their implications are discussed explicitly in the engineering-reliability section of this report.

Prepared as part of the Hossein Molhem Engineering Portfolio.

Abstract

This report presents the analytical design and CST full-wave simulation of a 5G microstrip patch antenna and a two-element phased patch array operating near 2.501 GHz. The design uses a DiClad 880-IM substrate with $\varepsilon_r = 2.20$, $\tan \delta = 0.0009$, and $h = 1.524$ mm. Initial patch, feed-line, and inset dimensions were obtained from the rectangular-patch transmission-line model and implemented as a parameterized CST model.

The final tuned single patch used a patch length of 38.40 mm and an inset depth of 14.11 mm. Its simulated resonance occurred at 2.50507 GHz with $S_{11,\min} \approx -39.87$ dB. The PEC baseline produced peak directivity of approximately 7.226 dBi, radiation efficiency of 96.3%, and total efficiency of 96.2%. A supplementary fixed-mesh copper model preserved the resonance, input match, directivity, and normalized pattern while reducing the radiation and total efficiencies to 92.46% and 92.38%, respectively. This comparison quantified the conductor-loss effect omitted by the PEC baseline.

The element was extended to a two-port 2×1 array with center-to-center spacing $d = 59.935$ mm. Because mutual coupling changed the isolated-element resonant behavior, the array required independent retuning. The final array used a patch length of 36.70 mm and an inset depth of 13.35 mm. Near the operating frequency, the simulated results were approximately $S_{11} = -11.93$ dB, $S_{22} = -15.20$ dB, and mutual coupling of about -15 dB.

Beam steering was demonstrated by combining the two CST port farfields with equal amplitudes and relative phases of 0° , -90° , and $+90^\circ$. The in-phase case produced a broadside pattern with displayed peak directivity of approximately 9.655 dBi. The phase-shifted cases produced simulated beam displacements of approximately 20° in opposite half-planes, compared with the ideal two-element array-factor prediction of 30° magnitude.

The study links closed-form synthesis, full-wave modeling, element tuning, conductor-loss sensitivity, two-port characterization, mutual-coupling interpretation, and phase-controlled beam steering. The results are analytical or simulated; no fabricated prototype or measured data are claimed. Formal mesh- and boundary-convergence studies were not completed because of the CST Learning Edition resource limit.

Project Objective

The objective of this project is to develop and evaluate a complete antenna-engineering workflow for a 5G microstrip patch antenna and a two-element phased patch array operating near 2.501 GHz.

The work integrates closed-form antenna design, parameterized full-wave electromagnetic simulation, impedance matching, radiation analysis, mutual-coupling characterization, and phase-controlled beam steering.

The project has four principal technical objectives:

1. Design an inset-fed rectangular microstrip patch antenna using analytical transmission-line relations for the patch dimensions, effective dielectric constant, fringing-field extension, microstrip-feed width, and initial inset position.
2. Implement and tune the single-patch model in CST Studio Suite, and evaluate its resonant frequency, reflection coefficient, directivity, radiation pattern, radiation efficiency, and total efficiency.
3. Extend the tuned element into a two-port 2×1 patch array, quantify the effect of mutual coupling, and determine the separate tuning required to maintain acceptable input matching near the operating frequency.
4. Demonstrate beam steering by applying controlled relative phase shifts between the two array elements and comparing the simulated scan angles with ideal two-element array-factor predictions.

A secondary objective is to evaluate the numerical reliability and physical limitations of the simulation model. Particular attention is given to the CST Learning Edition mesh-cell restriction, the use of PEC metallization in the baseline models, the supplementary finite-conductivity copper sensitivity study, and the absence of formal mesh- and boundary-convergence studies.

The final outcome is intended to provide a technically traceable portfolio record that connects analytical antenna theory, full-wave simulation, numerical interpretation, and phased-array behavior. All reported performance results are analytical or simulated; no fabricated prototype or measured antenna data are claimed.

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Chapter 1

5G Microstrip Patch Array Design and Beam Steering

1.1 Introduction

Fifth-generation wireless systems use multiple frequency ranges, and planar microstrip antennas are widely studied for compact sub-6-GHz radio-frequency front ends. Rectangular microstrip patch antennas are attractive because they are low profile, planar, mechanically simple, and compatible with printed-circuit fabrication. Their principal limitations include relatively narrow impedance bandwidth and sensitivity to substrate properties, feed geometry, conductor loss, and fabrication tolerances.

This report treats the antenna as a complete engineering design problem. The CST model is not used as a black box; instead, the geometry is initialized from microstrip antenna theory, tuned in full-wave simulation, and interpreted using performance metrics such as S_{11} , radiation pattern, directivity, radiation efficiency, and total efficiency. The tuned patch is then used as the starting point for a two-element array. The array analysis connects full-wave simulation to phased-array theory through element spacing, array factor, mutual coupling, and beam steering under controlled relative excitation phase.

1.2 Engineering Scope and Deliverables

This project develops a complete antenna-engineering workflow consisting of three connected technical stages: design and tuning of a single microstrip patch antenna, extension of the tuned element into a two-port 2×1 patch array, and demonstration of phase-controlled beam steering.

The analytical and simulation workflow includes the following tasks:

- determine the operating frequency and substrate parameters;
- calculate the initial rectangular-patch and inset-feed dimensions;
- construct a parameterized single-patch CST model;
- tune the resonant frequency and input impedance;
- evaluate reflection coefficient, radiation pattern, directivity, radiation efficiency, and total efficiency;
- compare PEC and finite-conductivity metallization models;
- construct and independently retune a two-port patch array;
- quantify input matching and mutual coupling;
- combine the individual port farfields using controlled amplitude-phase excitation; and
- evaluate broadside and steered radiation patterns.

The report emphasizes traceability between analytical calculations, CST model parameters, simulation results, and engineering interpretation. Key dimensions, material assumptions, solver settings, numerical limitations, and validation evidence are documented explicitly so that the design process can be independently reviewed and reproduced.

Table 1.1 summarizes the principal technical deliverables and their locations in this report.

1.3 Design Specifications and Material Selection

The public portfolio edition treats the operating frequency, substrate, and laminate thickness as fixed engineering design inputs. These parameters define the analytical patch synthesis, CST material model, frequency-domain simulation range, farfield-monitor frequency, and array-spacing calculations used throughout the project.

Table 1.1: Principal engineering deliverables and their report locations.

Engineering Deliverable	Report Location
Operating frequency, substrate, and material parameters	Section 1.3
Rectangular-patch analytical design	Sections 1.4–1.5
Single-patch CST construction and tuning	Sections 1.6–1.7
Finite-conductivity sensitivity analysis	Section 1.7.4
Two-element array theory and CST construction	Sections 1.8–1.9
Array matching and mutual-coupling analysis	Section 1.10
Beam-steering theory and CST implementation	Section 1.11
Numerical reliability and modeling limitations	Section 1.12.5
Engineering evidence matrix	Section 1.13

1.3.1 Operating Frequency

The selected design frequency is

$$f_0 = 2.501 \text{ GHz.}$$

This value is used consistently as the reference frequency for the analytical calculations, CST tuning process, farfield evaluation, and beam-steering analysis.

The corresponding free-space wavelength is

$$\lambda_0 = \frac{c}{f_0} = 119.869 \text{ mm,}$$

where c is the speed of light in free space.

1.3.2 Substrate and Laminate Thickness

The antenna is designed using DiClad 880-IM, a low-loss microwave laminate suitable for printed RF and antenna structures. The nominal dielectric properties adopted for the analytical calculations and CST material model are based on the manufacturer's published laminate data [1]:

$$\varepsilon_r = 2.20, \quad \tan \delta = 0.0009.$$

The selected laminate thickness is

$$h = 1.524 \text{ mm} = 0.060 \text{ in.}$$

The relative permittivity and loss tangent are treated as nominal design values. In a fabricated implementation, manufacturing tolerance, material-lot variation, copper

roughness, and frequency dependence of the laminate properties could introduce additional resonance and efficiency variation.

Table 1.2 summarizes the principal design inputs used throughout the report.

Table 1.2: Principal antenna design specifications and material parameters.

Design Parameter	Selected Value
Design frequency, f_0	2.501 GHz
Free-space wavelength, λ_0	119.869 mm
Substrate	DiClad 880-IM
Substrate thickness, h	1.524 mm (0.060 in)
Relative permittivity, ε_r	2.20
Loss tangent, $\tan \delta$	0.0009
Single-element feed topology	Inset-fed microstrip line
Nominal feed impedance	50 Ω
Array topology	Two-port 2×1 linear patch array
Initial array spacing	$d \approx \lambda_0/2 = 59.935$ mm

1.4 Microstrip Patch Antenna Theory

A rectangular microstrip patch antenna consists of a conducting patch printed on a thin dielectric substrate backed by a ground plane. In the dominant half-wave rectangular-patch mode, the patch is approximately resonant when its effective length is close to one half of the guided wavelength in the substrate. The physical length is slightly shorter than this half-wavelength value because fringing fields extend beyond the open patch edges. This transmission-line interpretation is consistent with the standard half-wavelength rectangular microstrip-antenna model, in which the patch behaves as a resonant, low-profile radiator with a broadside beam and typically narrow impedance bandwidth [2].

For the dominant TM_{10} -like mode, the length dimension controls the resonant frequency most strongly, while the width affects radiation conductance, input resistance, bandwidth, and efficiency. The transmission-line design model gives the initial patch width as

$$W = \frac{c}{2f_0} \sqrt{\frac{2}{\varepsilon_r + 1}}, \quad (1.1)$$

where c is the speed of light, f_0 is the target resonant frequency, and ε_r is the substrate relative permittivity.

The effective dielectric constant is approximated by

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-1/2}, \quad (1.2)$$

where h is the substrate thickness. The normalized fringing length extension is

$$\frac{\Delta L}{h} = 0.412 \frac{(\varepsilon_{\text{eff}} + 0.3)(W/h + 0.264)}{(\varepsilon_{\text{eff}} - 0.258)(W/h + 0.8)}. \quad (1.3)$$

The effective resonant length is

$$L_{\text{eff}} = \frac{c}{2f_0 \sqrt{\varepsilon_{\text{eff}}}}, \quad (1.4)$$

and the physical patch length is

$$L = L_{\text{eff}} - 2\Delta L. \quad (1.5)$$

1.4.1 Ground Plane and Substrate Size

A finite ground plane is required in CST, even though the simple analytical model assumes a large ground plane. A common first-pass engineering rule is to extend the ground plane and substrate by approximately $3h$ beyond the patch on each side, giving

$$W_g \approx W + 6h, \quad L_{g,\text{min}} \approx L + 6h. \quad (1.6)$$

For an inset-fed model, the feed line also requires physical board length. Therefore, the CST model uses a practical feed-line extension L_f and the corresponding substrate length estimate

$$L_g \approx L + L_f + 6h. \quad (1.7)$$

This value is a simulation starting point. The ground size can be increased later if edge diffraction or back radiation becomes excessive in the CST pattern.

1.4.2 Inset-Fed Microstrip-Line Matching Model

The selected feed topology for the initial single-patch CST model is an inset-fed microstrip line. This choice keeps the feed printed on the same substrate as the patch, is easy to document with top-view and cross-sectional CST screenshots, and provides a tunable impedance transformation by moving the feed point away from the radiating edge.

The $50 \, \Omega$ microstrip-feed width is obtained from the standard quasi-static microstrip-line relations. With

$$u = \frac{W_f}{h},$$

the effective dielectric constant of the feed line is approximated by

$$\varepsilon_{\text{eff},f} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left(1 + \frac{12}{u}\right)^{-1/2}. \quad (1.8)$$

For $u \leq 1$,

$$Z_0 = \frac{60}{\sqrt{\varepsilon_{\text{eff},f}}} \ln \left(\frac{8}{u} + \frac{u}{4} \right), \quad (1.9)$$

and for $u > 1$,

$$Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{\text{eff},f}} [u + 1.393 + 0.667 \ln(u + 1.444)]}. \quad (1.10)$$

The feed width W_f is then found by solving $Z_0 = 50 \, \Omega$.

The inset depth is estimated using the approximate input-resistance variation along the patch length,

$$R_{\text{in}}(y_0) \approx R_{\text{edge}} \cos^2 \left(\frac{\pi y_0}{L} \right), \quad (1.11)$$

where y_0 is measured inward from the radiating edge along the patch length. Solving for a target input resistance $R_0 = 50 \, \Omega$ gives

$$y_0 \approx \frac{L}{\pi} \cos^{-1} \left(\sqrt{\frac{R_0}{R_{\text{edge}}}} \right). \quad (1.12)$$

Because the edge resistance depends on the actual patch width, substrate, feed discontinuity, finite ground plane, and fringing fields, R_{edge} is treated as a first-pass engineering estimate. The initial script uses $R_{\text{edge}} = 300 \, \Omega$, then CST tuning will adjust the inset depth and inset gap to minimize S_{11} at f_0 .

For traceability, the numerical implementation of the patch, feed-line, ground-plane, and inset-depth calculations is included in Appendix A. The Python source is given in Listing A.1, and the corresponding console output is reported in Listing A.2. This connects the theoretical model directly to the initial parameter set used in the CST model.

1.5 Single Patch Analytical Design

Using the selected design frequency

$$f_0 = 2.501 \, \text{GHz},$$

the selected substrate thickness

$$h = 1.524 \, \text{mm},$$

and the Rogers DiClad 880 design dielectric constant

$$\varepsilon_r = 2.20,$$

the free-space wavelength is

$$\lambda_0 = \frac{c}{f_0} = 119.869 \text{ mm.} \quad (1.13)$$

Substituting these values into Eqs. (1.1)–(1.5) gives the analytical single-patch starting dimensions summarized in Table 1.3. The same calculation is verified independently by the Python utility in Appendix A; the printed console output in Listing A.2 is used as the numerical audit trail for the values reported in this section.

Table 1.3: Initial rectangular patch dimensions before CST tuning.

Quantity	Initial Value
Target frequency, f_0	2.501 GHz
Free-space wavelength, λ_0	119.869 mm
Substrate thickness, h	1.524 mm
Relative permittivity, ε_r	2.20
Patch width, W	47.382 mm
Effective permittivity, ε_{eff}	2.1097
Length extension, ΔL	0.803 mm
Effective length, L_{eff}	41.264 mm
Patch length, L	39.657 mm
Minimum ground width, $W_g \approx W + 6h$	56.526 mm
Minimum ground length, $L_{g,\text{min}} \approx L + 6h$	48.801 mm

1.5.1 Initial Inset-Fed CST Parameters

For the first CST model, an inset-fed microstrip line is selected. The feed-line width is calculated from the 50Ω microstrip-line model in Eqs. (1.8)–(1.10). For $\varepsilon_r = 2.20$ and $h = 1.524$ mm, the calculated feed width is

$$W_f = 4.735 \text{ mm.}$$

This value gives an approximate 50Ω printed feed line before the feed is connected to the patch.

The inset-depth estimate uses Eq. (1.12) with

$$R_{\text{edge}} = 300 \Omega, \quad R_0 = 50 \Omega.$$

This gives

$$y_0 = 14.520 \text{ mm}$$

measured inward from the radiating edge. Because the edge resistance is only an analytical estimate, y_0 should be treated as a CST tuning variable rather than a final manufactured dimension. The initial inset gap is selected as

$$g_i = 1.000 \text{ mm},$$

which provides a clear CST geometry gap on both sides of the inset feed while leaving room for later matching adjustment.

Table 1.4: Initial CST parameter set for the inset-fed single patch.

CST Parameter	Value	Unit	Purpose
f_0	2.501	GHz	Far-field monitor and target resonance
ε_r	2.20	–	DiClad 880 dielectric constant
$\tan \delta$	0.0009	–	Dielectric loss tangent
h	1.524	mm	Substrate thickness
W	47.382	mm	Patch width
L	39.657	mm	Patch length; main tuning variable
W_f	4.735	mm	Initial 50 Ω microstrip feed width
y_0	14.520	mm	Initial inset depth from patch edge
g_i	1.000	mm	Initial inset gap on each side of feed
L_f	15.000	mm	Feed-line extension for port setup
W_g	56.526	mm	Initial substrate/ground width
L_g	63.801	mm	Initial substrate/ground length including feed extension

The first CST simulation should use the values in Table 1.4. After the first S_{11} sweep, the patch length L , inset depth y_0 , and inset gap g_i will be tuned. The primary resonance-control variable is L ; increasing L generally shifts the resonance downward, while decreasing L shifts it upward. The primary impedance-control variables are y_0 and g_i . This separation of tuning variables will make the CST tuning process more systematic and easier to explain in the final engineering discussion.

1.6 CST Model Construction for the Single Patch

The single-patch CST model was constructed as a parameterized inset-fed microstrip antenna using the analytical dimensions developed in Section 1.5. The implemented model contained a finite ground plane, a DiClad 880-IM substrate, finite-thickness top and bottom metallization, an inset microstrip feed, and a waveguide port located at

the feed-line boundary. The parameterized construction allowed the patch length and inset depth to be varied without rebuilding the geometry.

1.6.1 Implemented CST Parameter Set

Table 1.5 lists the principal parameter names used in the CST model. The names `lossTan` and `h` match the implemented CST project. The initial values were used for the analytical starting model; `patchL` and `insetY` were subsequently changed during tuning.

Table 1.5: Principal CST parameters used for the single-patch model.

Parameter	Initial Value	Unit	Function
<code>f0</code>	2.501	GHz	Target frequency and farfield-monitor frequency
<code>fmin, fmax</code>	2.0, 3.0	GHz	Frequency-domain simulation range
<code>epsr</code>	2.20	–	Relative permittivity of DiClad 880-IM
<code>lossTan</code>	0.0009	–	Dielectric loss tangent
<code>h</code>	1.524	mm	Substrate thickness
<code>copperT</code>	0.035	mm	Physical metallization thickness used in the geometry
<code>copperSigma</code>	5.87×10^7	S/m	Copper conductivity used in the initial lossy-metal trial
<code>patchW</code>	47.382	mm	Patch width in the CST x -direction
<code>patchL</code>	39.657	mm	Initial patch length; principal resonance-tuning variable
<code>feedW</code>	4.735	mm	Approximate $50\ \Omega$ microstrip-feed width
<code>feedL</code>	15.000	mm	Distance from the board edge to the patch edge
<code>insetY</code>	14.520	mm	Initial inset depth measured along the patch length
<code>insetGap</code>	1.000	mm	Gap on each side of the inset feed
<code>gndW</code>	56.526	mm	Ground-plane and substrate width
<code>gndL</code>	63.801	mm	Ground-plane and substrate length
<code>arrayD</code>	59.935	mm	Element spacing reserved for the later 2×1 array
<code>portW</code>	<code>3 feedW</code>	mm	Waveguide-port width
<code>portH</code>	<code>5 h</code>	mm	Waveguide-port height

Additional derived parameters were used to keep the brick coordinates readable, including `halfGndW`, `halfGndL`, `halfPatchW`, `halfFeedW`, `slotHalfW`, `zTopSub`, `zTopMetal`, `patchYmin`, `patchYmax`, `insetYend`, `feedYmin`, and `feedYmax`.

1.6.2 Implemented Coordinate Convention and Layer Stack

The coordinate convention used in the actual CST model was

- the patch width and later array axis were aligned with the CST x -axis;

- the patch length and feed-line direction were aligned with the CST y -axis;
- the substrate thickness was aligned with the CST z -axis;
- the feed entered the patch from the negative- y side; and
- the waveguide port was placed on the negative- y board boundary and oriented in the positive- y direction.

The finite-thickness layer stack was implemented as

$$0 \leq z \leq \text{copperT}$$

for the ground metallization,

$$\text{copperT} \leq z \leq \text{copperT} + h$$

for the DiClad 880-IM substrate, and

$$\text{copperT} + h \leq z \leq 2 \text{copperT} + h$$

for the patch and feed metallization. Numerically, the top of the substrate and top of the metal were

$$z_{\text{TopSub}} = 1.559 \text{ mm}, \quad z_{\text{TopMetal}} = 1.594 \text{ mm}.$$

1.6.3 Parameterized Brick Construction

The inset-fed top conductor was built using four connected rectangular bricks. This avoided a Boolean subtraction and made the inset geometry easier to tune. Table 1.6 records the coordinate logic used in the CST model.

Table 1.6: Coordinate logic used to construct the parameterized single-patch CST model.

Object	x range	y range	z range	Material or role
Ground	$-\text{halfGndW}$ to $+\text{halfGndW}$	$-\text{halfGndL}$ to $+\text{halfGndL}$	0 to copperT	Copper trial; PEC in final tuned run
Substrate	$-\text{halfGndW}$ to $+\text{halfGndW}$	$-\text{halfGndL}$ to $+\text{halfGndL}$	copperT to zTopSub	DiClad 880-IM
Feed line	$-\text{halfFeedW}$ to $+\text{halfFeedW}$	feedYmin to insetYend	zTopSub to zTopMetal	Top metallization
Patch main	$-\text{halfPatchW}$ to $+\text{halfPatchW}$	insetYend to patchYmax	zTopSub to zTopMetal	Main patch region
Left inset arm	$-\text{halfPatchW}$ to $-\text{slotHalfW}$	patchYmin to insetYend	zTopSub to zTopMetal	Left side of inset
Right inset arm	slotHalfW to halfPatchW	patchYmin to insetYend	zTopSub to zTopMetal	Right side of inset
Waveguide Port 1	$-\text{portW}/2$ to $+\text{portW}/2$	$y = \text{feedYmin}$	0 to portH	Normal to $+y$, one mode

The principal derived coordinates were

$$\text{feedYmin} = -\frac{\text{gndL}}{2}, \quad \text{patchYmin} = \text{feedYmin} + \text{feedL},$$

$$\text{patchYmax} = \text{patchYmin} + \text{patchL}, \quad \text{insetYend} = \text{patchYmin} + \text{insetY},$$

and

$$\text{slotHalfW} = \frac{\text{feedW}}{2} + \text{insetGap}.$$

For the initial analytical model, these relations gave

$$\text{feedYmin} = -31.9005 \text{ mm}, \quad \text{patchYmin} = -16.9005 \text{ mm},$$

$$\text{patchYmax} = 22.7565 \text{ mm}, \quad \text{insetYend} = -2.3805 \text{ mm}.$$

The final tuned single patch retained the same board, feed width, and inset gap but used $\text{patchL} = 38.40 \text{ mm}$ and $\text{insetY} = 14.11 \text{ mm}$.

The parameter and coordinate records are also stored in the following project files:

- `cst_setup/single_patch_parameters.csv`;
- `cst_setup/single_patch_coordinate_table.csv`; and
- `cst_setup/single_patch_build_sheet.md`.

1.6.4 Implemented Boundary, Port, and Solver Setup

The model was solved from 2.0 to 3.0 GHz using the frequency-domain general-purpose solver with a tetrahedral mesh. Port 1 was defined at $y = \text{feedYmin}$ with one mode, positive- y orientation, width $\text{portW} = 3 \text{ feedW}$, and height $\text{portH} = 5h$. The S -parameters were normalized to 50Ω . Open boundaries with added space were applied on all six sides, and a farfield monitor was placed at $f_0 = 2.501$ GHz.

The initial finite-conductivity copper trial exceeded the CST Learning Edition mesh-cell limit. The successful tuned runs therefore used PEC for the ground, patch, and feed while retaining the dielectric loss tangent of the substrate. The solver accuracy was set to 10^{-3} for the tuning workflow, and adaptive tetrahedral mesh refinement was disabled to remain within the available mesh limit. These modeling decisions and their implications are discussed in Section 1.12.5.

1.6.5 Required Geometry Documentation

The implemented geometry is documented in Section 1.7 using a clean CST top-view export, a layered cross-sectional perspective, a final dimensional table, and an explicit layer-stack table. This format avoids an unreadable full parameter-list overlay while preserving all dimensions required to reproduce the model. The cross-sectional documentation identifies the 0.035 mm bottom metallization, the 1.524 mm DiClad 880-IM substrate, and the 0.035 mm top metallization.

1.7 Single Patch Simulation Results

The single inset-fed patch was simulated first because it is the radiating element used later in the 2×1 array. The analytical dimensions in Section 1.5 were used as the CST starting point, and the model was then tuned using the patch length and inset depth. The single-patch CST simulations were performed over the frequency range 2.0 to 3.0 GHz. The final tuned single-patch model used PEC metallization to satisfy the CST Learning Edition mesh-cell limitation, while the dielectric substrate loss of DiClad 880-IM was retained through the specified loss tangent.

1.7.1 Final Single-Patch Geometry Documentation

The final tuned geometry is documented by the top-view CST export in Figure 1.1 and the layered cross-sectional perspective in Figure 1.2. Because the complete CST parameter list was longer than a single readable screen capture, the principal final

dimensions are reported explicitly in Table 1.7. This combination preserves a clear CAD view while providing an auditable dimensional record.

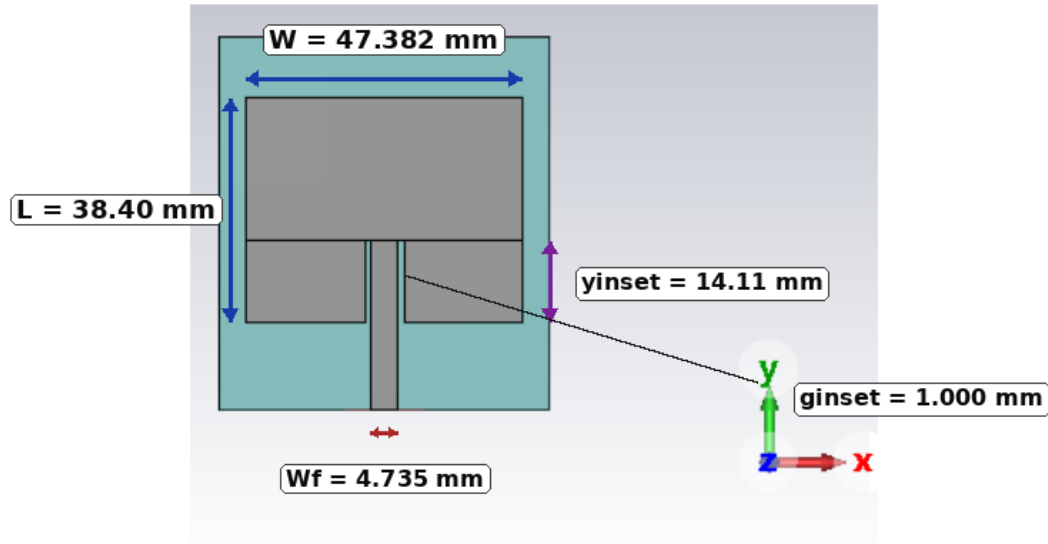


Figure 1.1: Top view of the final tuned inset-fed single-patch antenna. The patch width is aligned with the CST x -axis, and the patch length and feed line are aligned with the y -axis. The final dimensions are summarized in Table 1.7.

Table 1.7: Principal final CST geometry and material parameters of the tuned single-patch antenna.

Quantity	CST parameter	Final value	Function
Target frequency	f0	2.501 GHz	Selected operating frequency
Relative permittivity	epsr	2.20	DiClad 880-IM design value
Loss tangent	lossTan	0.0009	Dielectric loss retained in CST
Substrate thickness	h	1.524 mm	Dielectric-layer thickness
Metal thickness	copperT	0.035 mm	Geometric thickness of each metal layer
Patch width	patchW	47.382 mm	Dimension along the x -axis
Patch length	patchL	38.40 mm	Final resonant dimension along the y -axis
Feed-line width	feedW	4.735 mm	Approximate 50 Ω microstrip width
Feed-line length	feedL	15.000 mm	Feed extension from board edge to patch
Inset depth	insetY	14.11 mm	Final tuned feed position
Inset gap	insetGap	1.000 mm	Gap between feed and inset arms
Ground/substrate width	gndW	56.526 mm	Finite board width
Ground/substrate length	gndL	63.801 mm	Finite board length including feed extension

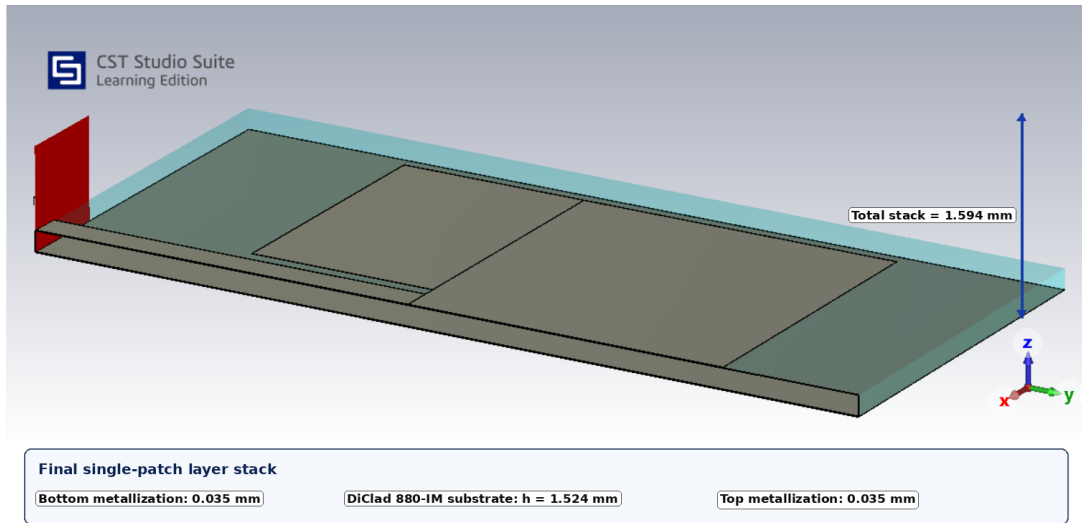


Figure 1.2: Layered cross-sectional perspective of the final single-patch CST model. The view shows the bottom ground metallization, DiClad 880-IM substrate, and top patch/feed metallization. The exact layer coordinates and thicknesses are listed in Table 1.8.

Table 1.8: Layer stack used in the final single-patch CST model.

Layer	$z_{\min}$ (mm)	$z_{\max}$ (mm)	Thickness (mm)
Ground metallization	0.000	0.035	0.035
DiClad 880-IM substrate	0.035	1.559	1.524
Top metallization	1.559	1.594	0.035

1.7.2 Single Patch Tuning Log

The patch length controls the resonant frequency, while the inset depth controls the input impedance match. The first simulation resonated below the target frequency, indicating that the patch was electrically too long. Therefore, the patch length was reduced and the inset depth was scaled to preserve a similar feed position. A final fine-tuning step moved the resonance very close to the selected design frequency.

Table 1.9: Single-patch CST tuning log.

Run	Geometry	S_{11} minimum	Observation
Initial analytical model	$L = 39.657$ mm, $y_{\text{inset}} = 14.520$ mm, $g_{\text{inset}} = 1.000$ mm	-20.39 dB at 2.431 GHz	The resonance was below the target frequency, indicating that the patch was electrically too long.
Retuned run	$L = 38.55$ mm, $y_{\text{inset}} = 14.11$ mm, $g_{\text{inset}} = 1.000$ mm	-30.70 dB at 2.491 GHz	The resonance moved close to the target frequency $f_0 = 2.501$ GHz.
Final single patch	$L = 38.40$ mm, $y_{\text{inset}} = 14.11$ mm, $g_{\text{inset}} = 1.000$ mm	-39.87 dB at 2.505 GHz	This final tuned result was selected for the single-patch radiation-pattern and efficiency analysis.

Figure 1.3 shows the three tuning curves, and Figure 1.4 isolates the final tuned result. The final resonant frequency is 2.50507 GHz, which is only about 4.07 MHz above the selected 2.501 GHz design target. This corresponds to a frequency error of approximately 0.16%, which is acceptable for the tuned CST model.

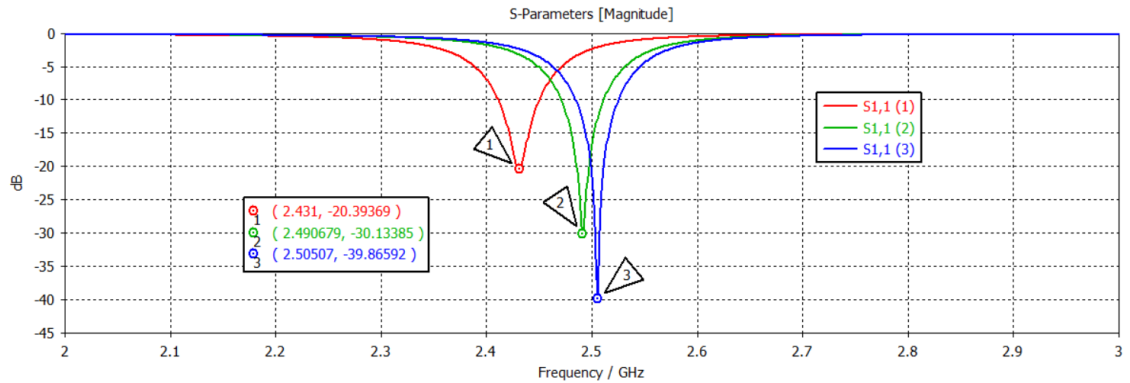


Figure 1.3: Single-patch S_{11} tuning history. The initial analytical design resonated at approximately 2.431 GHz. After reducing the patch length, the resonance moved toward the target frequency, and the final tuned run resonated near 2.505 GHz.

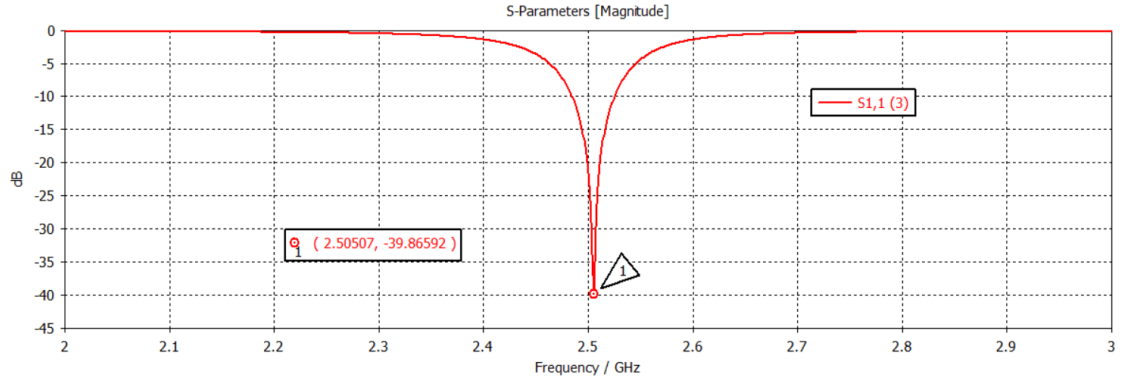


Figure 1.4: Final tuned reflection coefficient of the single inset-fed patch antenna. The final model resonated at approximately 2.50507 GHz with $S_{11,\min} \approx -39.87$ dB, very close to the selected design frequency of 2.501 GHz.

1.7.3 Single Patch Radiation Patterns and Efficiency

The final single-patch farfield result is shown in Figure 1.5. CST reported a maximum directivity of approximately 7.226 dBi at 2.501 GHz. The radiation efficiency and total efficiency were reported as -0.1634 dB and -0.1700 dB, respectively, corresponding to approximately 96.3% and 96.2%. Because PEC metallization was used for the tuned run, the radiation efficiency includes the modeled dielectric loss but excludes conductor loss and input-mismatch loss. The total efficiency additionally includes input-mismatch loss. Neither value includes finite-copper ohmic loss.

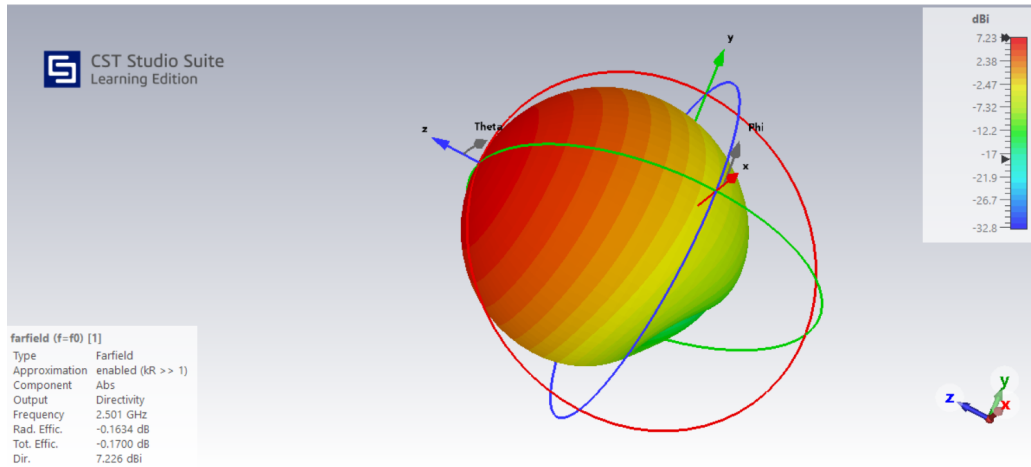


Figure 1.5: Three-dimensional CST farfield pattern of the final tuned single patch at 2.501 GHz. The reported maximum directivity is approximately 7.226 dBi, with radiation and total efficiencies of approximately 96.3% and 96.2%, respectively.

Figures 1.6 and 1.7 show the two principal 2D farfield cuts. The $\phi = 0^\circ$ cut gives a

main-lobe magnitude of approximately 7.23 dBi, a broadside main-lobe direction of 0° , and a 3-dB angular width of approximately 86.8° . These values are consistent with the expected broadside radiation of a rectangular microstrip patch.

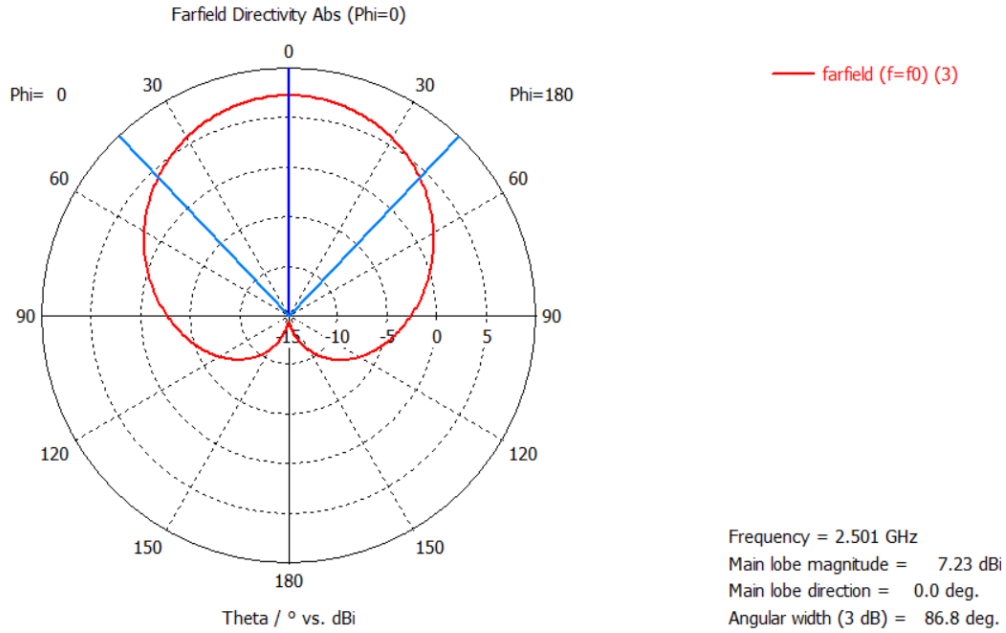


Figure 1.6: Single-patch farfield directivity cut for $\phi = 0^\circ$. The main lobe is broadside with a magnitude of about 7.23 dBi and an angular width of about 86.8° .

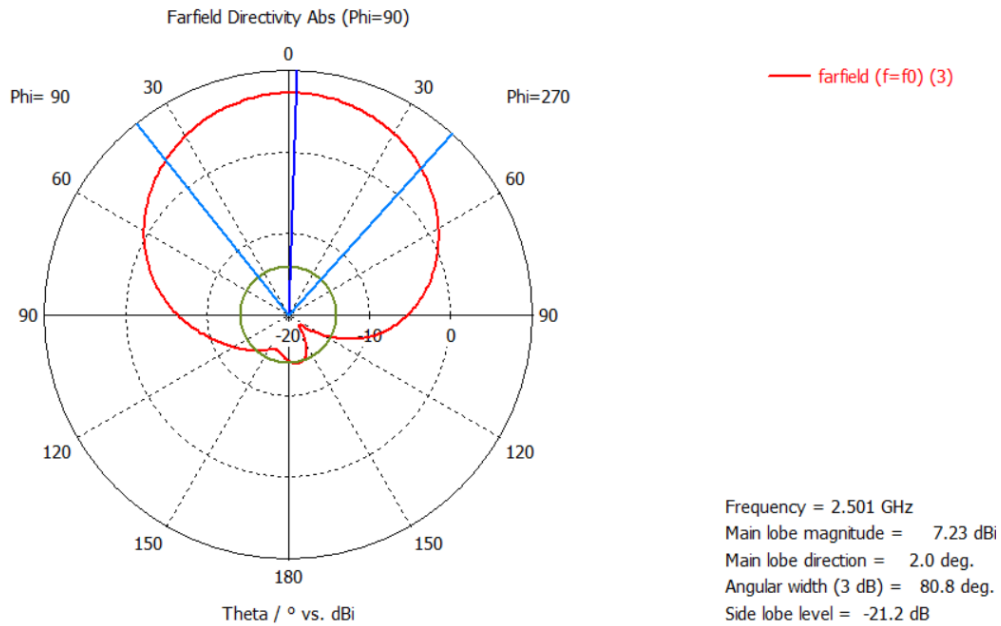


Figure 1.7: Single-patch farfield directivity cut for $\phi = 90^\circ$. This cut provides the orthogonal principal-plane response of the tuned patch element.

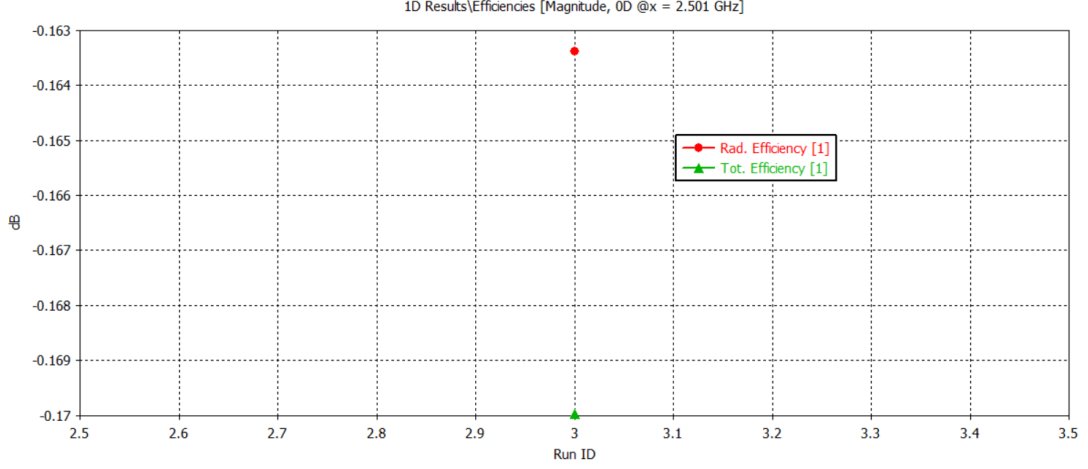


Figure 1.8: CST efficiency result for the final tuned single-patch PEC model at 2.501 GHz. The simulated radiation efficiency is approximately -0.1634 dB $\approx 96.3\%$, and the simulated total efficiency is approximately -0.1700 dB $\approx 96.2\%$.

Table 1.10: Single patch final simulation summary.

Metric	Final CST Value
Target frequency	2.501 GHz
Final resonant frequency	2.50507 GHz
Frequency error	0.16%
Minimum S_{11}	-39.87 dB
Radiation efficiency	-0.1634 dB $\approx 96.3\%$
Total efficiency	-0.1700 dB $\approx 96.2\%$
Directivity	7.226 dBi
Main-beam direction, $\phi = 0^\circ$ cut	0°
3-dB angular width, $\phi = 0^\circ$ cut	86.8°
Final tuned <code>patchL</code>	38.40 mm
Final tuned <code>insetY</code>	14.11 mm

1.7.4 Finite-Conductivity Copper Validation

A supplementary conductor-loss validation was performed on the final single-patch geometry after the PEC baseline had been completed. The geometry, substrate, port, boundaries, solver type, frequency range, and tuned dimensions were held fixed. The only physical change was replacement of the PEC metallization by the CST *Lossy metal* model with

$$\sigma_{\text{Cu}} = 5.87 \times 10^7 \text{ S/m.}$$

This produced a controlled comparison of the PEC approximation against a finite-conductivity copper model.

An initial lossy-metal run with adaptive tetrahedral mesh refinement was attempted. The adaptive sequence reached the fourth refinement pass, but the mesh then exceeded the 20,000-cell limit of the CST Learning Edition before a converged solution was obtained. The incomplete adaptive result was discarded. The simulation was then repeated using the same geometry and fixed initial-mesh settings, with mesh adaptation disabled. This fixed-mesh run completed successfully after eight frequency samples and produced valid S -parameters, impedance matrices, VSWR data, efficiencies, and farfield results. Consequently, the study is treated as a fixed-mesh conductor-model sensitivity check rather than a formal mesh-convergence study.

Figure 1.9 shows that finite copper conductivity had very little effect on the tuned resonance and impedance match. The lossy-metal model resonated near 2.504 GHz with $S_{11,\min} \approx -39.72$ dB, compared with 2.50507 GHz and -39.87 dB for the PEC baseline.

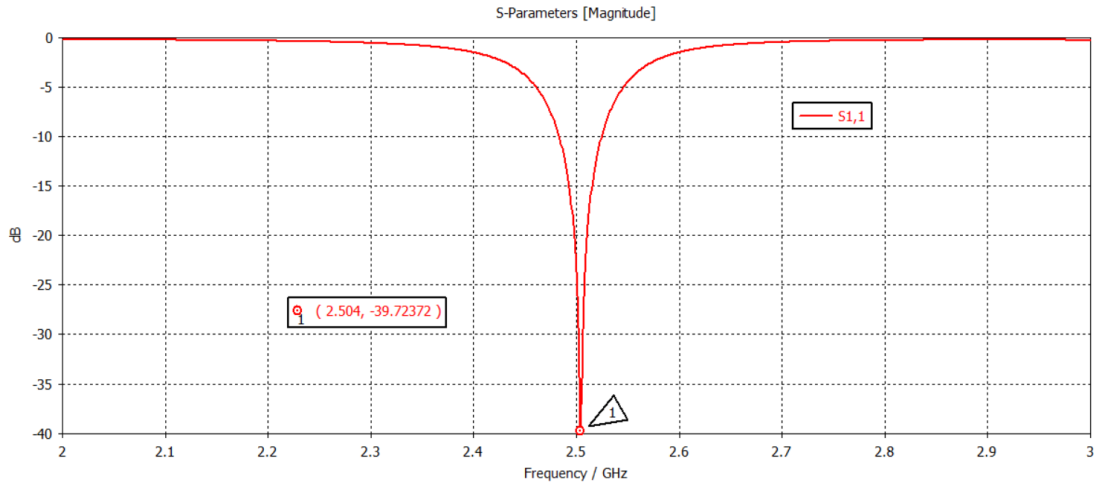


Figure 1.9: Reflection coefficient of the fixed-mesh finite-conductivity copper model. The resonance occurs near 2.504 GHz with $S_{11,\min} \approx -39.72$ dB, closely matching the PEC baseline.

The finite-conductivity model produced a radiation efficiency of -0.340329 dB, corresponding to approximately 92.46%, and a total efficiency of -0.344040 dB, corresponding to approximately 92.38%. The small difference between the two efficiencies is consistent with the very strong impedance match near the operating frequency. The efficiency reduction relative to PEC quantifies the conductor-loss contribution that was omitted from the baseline model.

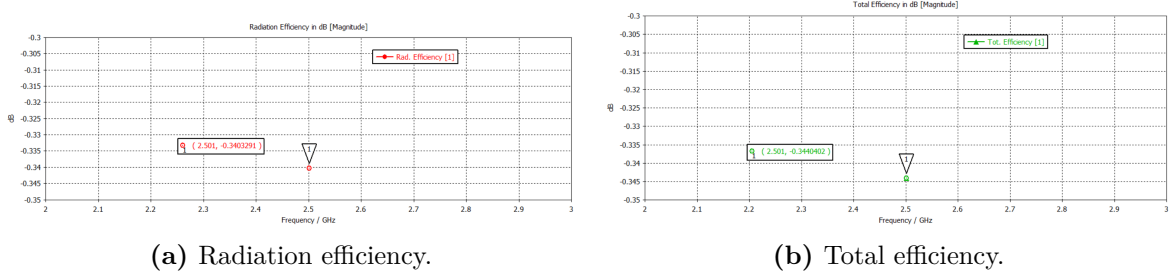


Figure 1.10: Efficiencies of the fixed-mesh finite-conductivity copper model at 2.501 GHz. The radiation efficiency is approximately 92.46%, and the total efficiency is approximately 92.38%.

The peak directivity remained approximately 7.23 dBi, essentially unchanged from the PEC value of 7.226 dBi. The $\phi = 90^\circ$ principal-plane cut retained a broadside main lobe near 2° , a 3-dB beamwidth of approximately 80.8° , and a sidelobe level of approximately -21.2 dB. The farfield comparison therefore indicates that conductor loss reduced efficiency without materially changing the normalized radiation-pattern shape or directivity.

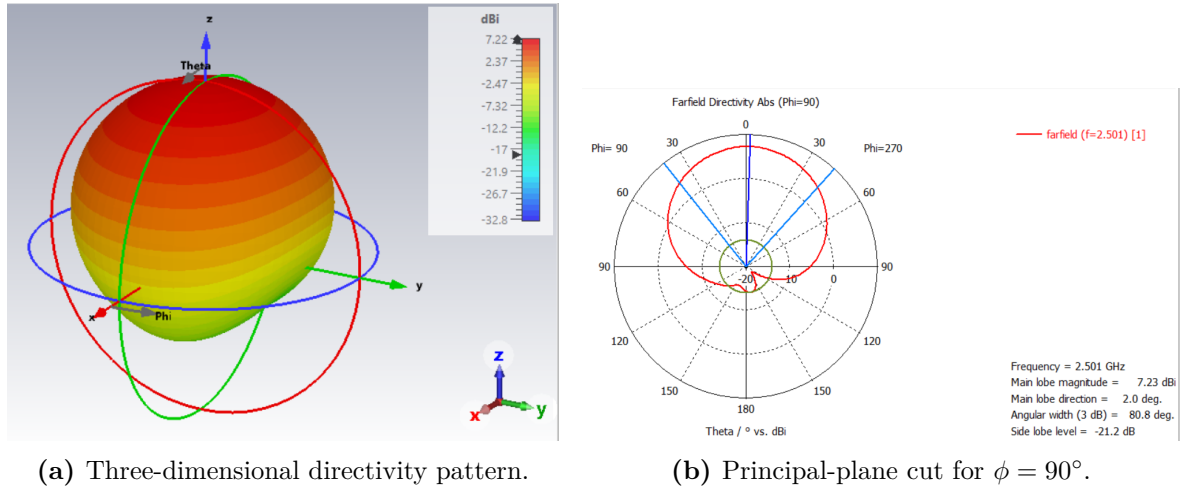


Figure 1.11: Radiation patterns of the fixed-mesh finite-conductivity copper model at 2.501 GHz. The peak directivity remains approximately 7.23 dBi, and the principal-plane pattern remains consistent with the PEC result.

Table 1.11: PEC and finite-conductivity copper comparison for the final single-patch geometry.

Metric	PEC baseline	Copper lossy metal	Engineering interpretation
Resonant frequency	2.50507 GHz	2.504 GHz	Change of approximately -1.07 MHz
Minimum S_{11}	-39.87 dB	-39.72 dB	Matching essentially unchanged
Peak directivity	7.226 dBi	≈ 7.23 dBi	Directivity essentially unchanged
Radiation efficiency	96.3%	92.46%	Reduction of approximately 3.84 percentage points
Total efficiency	96.2%	92.38%	Reduction of approximately 3.78 percentage points
$\phi = 90^\circ$ 3-dB beamwidth	80.8°	80.8°	Normalized pattern remained stable

The comparison confirms that the PEC approximation was accurate for resonant frequency, impedance matching, directivity, and radiation-pattern shape in this project. Its principal limitation was an overestimate of radiation and total efficiency because finite copper ohmic loss was excluded.

1.8 Two-Element Array Theory and Design

A 2×1 patch array increases directivity by combining the fields from two patch elements. If mutual coupling and element-pattern distortion are neglected, the total pattern can be approximated by pattern multiplication:

$$F_{\text{total}}(\theta, \phi) \approx F_{\text{element}}(\theta, \phi)AF(\theta, \phi),$$

where F_{element} is the single-patch element pattern and AF is the array factor.

For two identical elements along the x -axis with spacing d , equal amplitudes, and progressive phase shift $\Delta\phi$, the normalized array factor in the array plane can be written as

$$AF(\theta) = 2 \cos \left[\frac{1}{2} (k_0 d \sin \theta + \Delta\phi) \right],$$

where $k_0 = 2\pi/\lambda_0$. A broadside beam occurs for $\Delta\phi = 0$. Beam steering to an angle

θ_0 is obtained approximately from

$$\Delta\phi = -k_0 d \sin \theta_0.$$

The recommended starting center-to-center element spacing is

$$d \approx \frac{\lambda_0}{2} \approx 59.94 \text{ mm}$$

for $f_0 = 2.501$ GHz. In practice, the spacing may need adjustment to reduce coupling, avoid grating lobes, and fit the physical patch and feed geometry. The phase-shift values used as beam-steering starting points are verified using the Python calculation in Appendix B, with the console output documented in Listing B.2.

For a two-element array with $d = \lambda_0/2$, the ideal progressive phase shift required to steer the main beam toward θ_0 is

$$\Delta\phi = -k_0 d \sin \theta_0 = -180^\circ \sin \theta_0.$$

The resulting initial phase-shift values are summarized in Table 1.12.

Table 1.12: Initial progressive phase shifts for beam steering with $d = \lambda_0/2$.

Target angle, θ_0	Phase-shift formula	Progressive phase, $\Delta\phi$
0°	$-180^\circ \sin(0^\circ)$	0°
15°	$-180^\circ \sin(15^\circ)$	-46.6°
30°	$-180^\circ \sin(30^\circ)$	-90.0°
45°	$-180^\circ \sin(45^\circ)$	-127.3°

1.9 CST Model Construction for the 2×1 Patch Array

The 2×1 array model was constructed from the tuned single-patch element. The final single-patch geometry was duplicated in the x -direction, and the center-to-center spacing was set to

$$d = 59.935 \text{ mm} \approx \frac{\lambda_0}{2}.$$

The substrate and ground plane were extended so that both elements and both feed lines were fully supported. Each element was fed by its own waveguide port so that the two ports could be excited independently for matching, coupling, and beam-steering analysis.

The same mesh-control simplification used for the final single-patch run was retained: the metal regions were modeled as PEC, while the DiClad 880-IM dielectric loss was retained. This approach kept the CST model below the mesh-cell limitation while preserving the dominant resonant and radiation behavior needed for the antenna and array analysis.

1.9.1 Final Array Geometry Parameters

Table 1.13: Final tuned CST dimensions for the 2×1 array model.

Parameter	Final Value
Target frequency, f_0	2.501 GHz
Substrate	DiClad 880-IM
Substrate thickness, h	1.524 mm
Patch width, patchW	47.382 mm
Final array patch length, patchL	36.70 mm
Feed width, feedW	4.735 mm
Feed length, feedL	15.000 mm
Inset depth, insetY	13.35 mm
Inset gap, insetGap	1.000 mm
Element spacing, arrayD	59.935 mm
Metallization model for tuned run	PEC approximation

1.9.2 CST Array Construction Notes

The first array run used the final tuned single-patch dimensions directly. The array resonance then shifted downward because the second element changed the electromagnetic environment of the first element. This behavior is expected in a closely spaced array because the input impedance of each radiating element is affected by mutual coupling. Therefore, the array required a separate tuning sequence after the single-patch tuning was completed.

The final CST array model used two waveguide ports. Port 1 excited the first element and Port 2 excited the duplicated element. The two-port model allowed CST to compute S_{11} , S_{22} , S_{12} , and S_{21} , which were used to document input matching and mutual coupling.

1.10 Two-Element Array Simulation Results

The tuned single-patch element was duplicated to form the 2×1 array. After duplication, the array did not retain the exact same resonance as the isolated single element because the two elements were electromagnetically coupled. Therefore, the array was tuned separately. The final array tuning used $\text{patchL} = 36.70$ mm and $\text{insetY} = 13.35$ mm, while preserving $\text{patchW} = 47.382$ mm, $\text{feedW} = 4.735$ mm, and the element spacing $d = 59.935$ mm.

1.10.1 Array Geometry

Figures 1.12 and 1.13 document the final two-element geometry and layer stack. The two elements were arranged along the CST x -axis with center-to-center spacing $d = 59.935$ mm, while both feed lines were aligned with the y -axis. The complete tuned dimensional set is reported in Table 1.13; therefore, the figures can remain visually clear without a crowded parameter-list overlay.

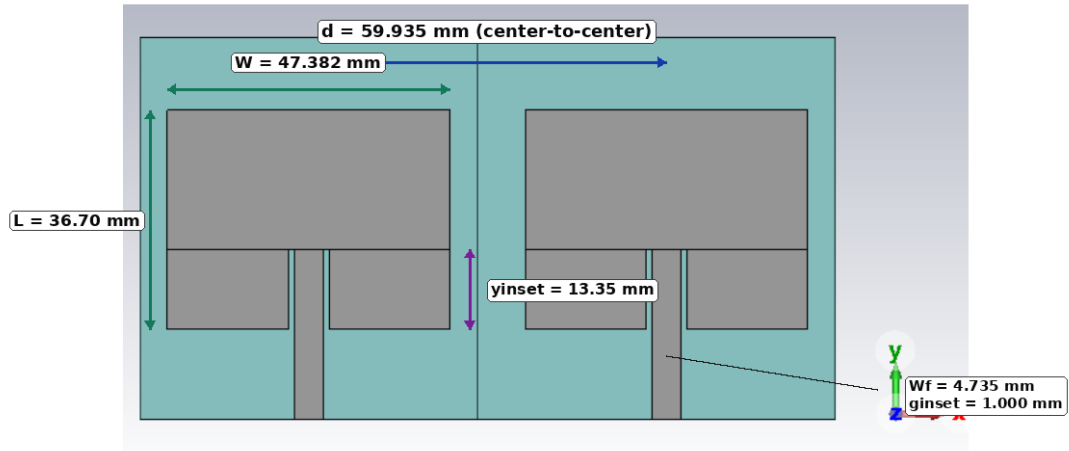


Figure 1.12: Top view of the final tuned 2×1 inset-fed patch array. The final elements used $W = 47.382$ mm, $L = 36.70$ mm, $y_{\text{inset}} = 13.35$ mm, $g_{\text{inset}} = 1.000$ mm, and center-to-center spacing $d = 59.935$ mm.

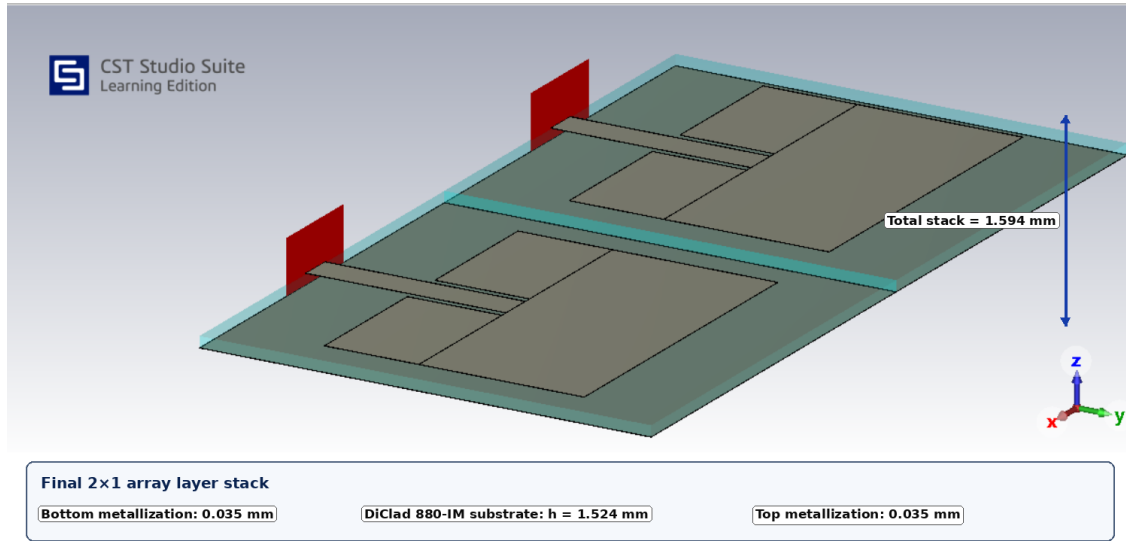


Figure 1.13: Layered cross-sectional perspective of the final 2×1 array. Both elements use the same 0.035 mm bottom metallization, 1.524 mm DiClad 880-IM substrate, and 0.035 mm top metallization stack.

Table 1.14: Layer stack used in the final 2×1 array model.

Layer	$z_{\min}$ (mm)	$z_{\max}$ (mm)	Thickness (mm)
Ground metallization	0.000	0.035	0.035
DiClad 880-IM substrate	0.035	1.559	1.524
Top metallization	1.559	1.594	0.035

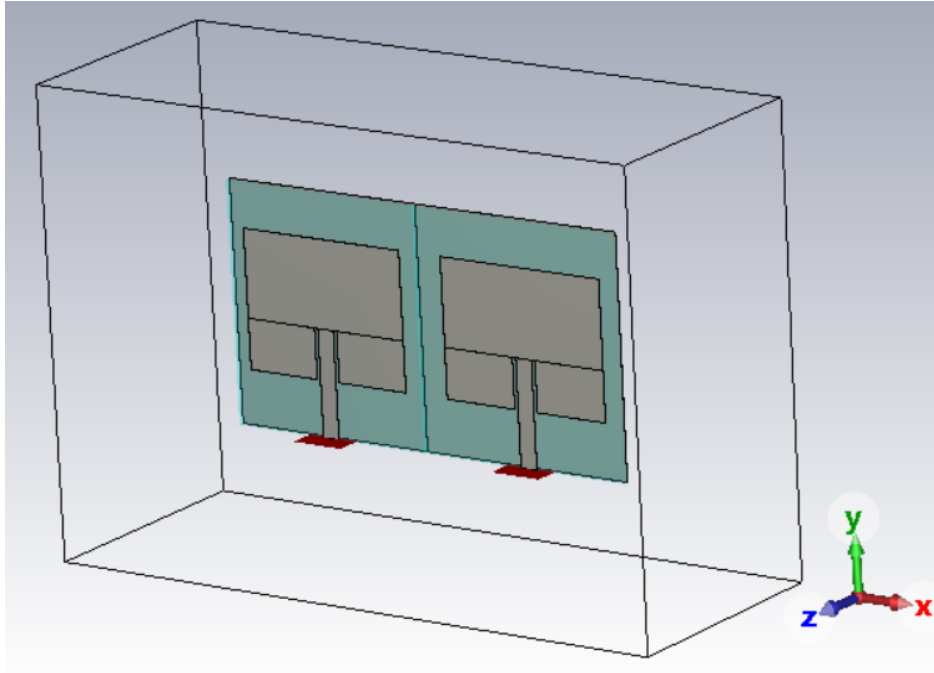


Figure 1.14: Two-port CST model of the 2×1 array. Each patch element is fed by a separate waveguide port, enabling both two-port S -parameter extraction and independent phase excitation for beam steering.

1.10.2 Array Input Match and Mutual Coupling

The first array simulation using the single-patch tuned length showed a downward resonant-frequency shift. This is a direct consequence of mutual coupling and the changed boundary environment when a second element is introduced. Table 1.15 summarizes the array tuning steps. The final Run 05 represents the best compromise between input matching, resonance near the target frequency, and mutual coupling.

Table 1.15: Two-element array tuning log.

Run	Geometry	S -parameter behavior near target	Coupling	Observation
Array Run 01	$L = 38.40$ mm, $y_{\text{inset}} = 14.11$ mm	Resonance near 2.399 GHz.	≈ -13.7 dB	The resonance shifted downward after forming the two-element array.
Array Run 02	$L = 36.85$ mm, $y_{\text{inset}} = 13.54$ mm	$S_{11} \approx -16.9$ dB, $S_{22} \approx -17.8$ dB.	≈ -14.7 dB	The matching improved, but the resonance remained slightly below the target frequency.
Array Run 03/04	$L = 36.65$ mm, $y_{\text{inset}} = 13.54/13.20$ mm	Improved frequency alignment; port balance varied.	≈ -15 dB	These runs were used to study the sensitivity of the array response to the feed inset position.
Final Run 05	$L = 36.70$ mm, $y_{\text{inset}} = 13.35$ mm	$S_{11} \approx -11.9$ dB, $S_{22} \approx -15.2$ dB.	≈ -15 dB	This was selected as the final tuned array for the beam-steering simulations.

Figures 1.15 and 1.16 show the final two-port S -parameters for Run 05. Near the operating frequency, the array satisfies the common -10 dB input-match criterion for both ports. The mutual coupling is approximately -15 dB, which is an important array-level result because it explains why the array had to be retuned separately from the isolated element.

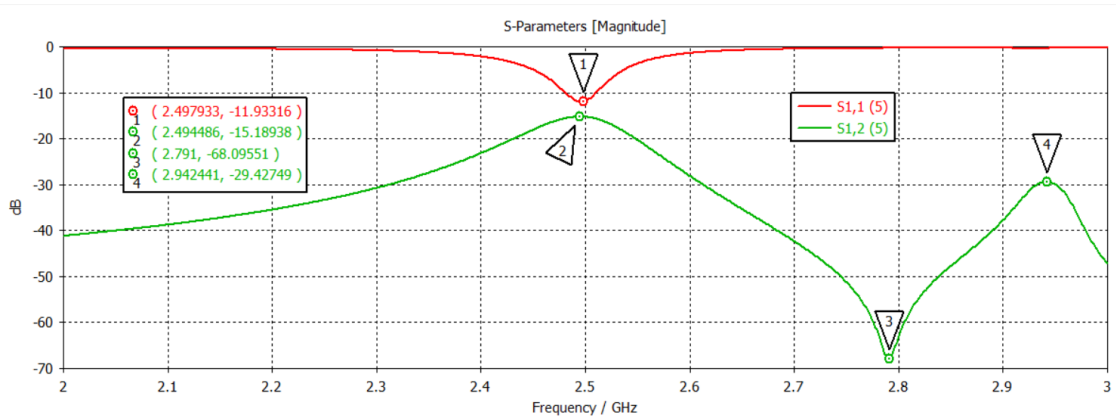


Figure 1.15: Final Run 05 S_{11} and S_{12} for the tuned 2×1 array. The final array achieved $S_{11} \approx -11.93$ dB near 2.498 GHz and mutual coupling $S_{12} \approx -15.19$ dB near 2.494 GHz.

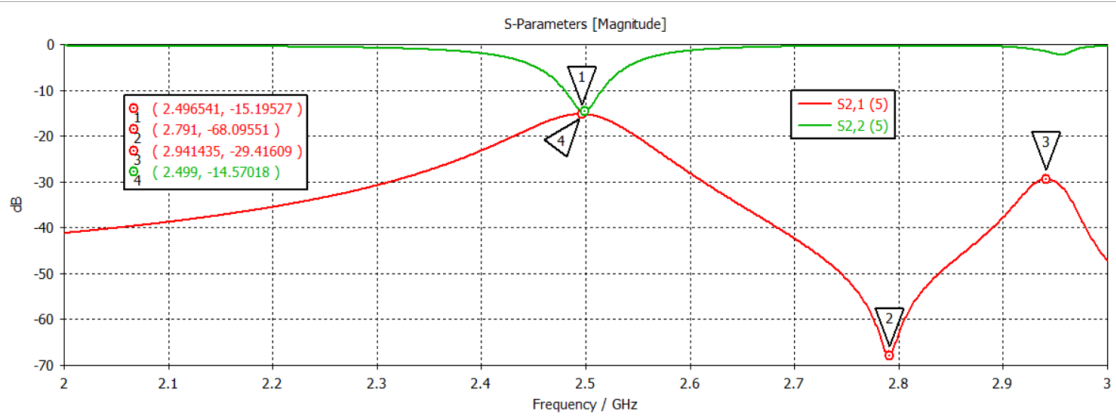


Figure 1.16: Final Run 05 S_{21} and S_{22} for the tuned 2×1 array. The second port achieved $S_{22} \approx -15.20$ dB near 2.497 GHz, and the coupling from Port 1 to Port 2 was approximately $S_{21} \approx -14.57$ dB near 2.499 GHz.

1.10.3 Individual Element Farfield Responses

Before combining the port excitations for beam steering, CST produced farfield responses for each individual port excitation. These responses were used to verify that each port excited a radiating patch element and to provide the element-level data used by the post-processing combination step.

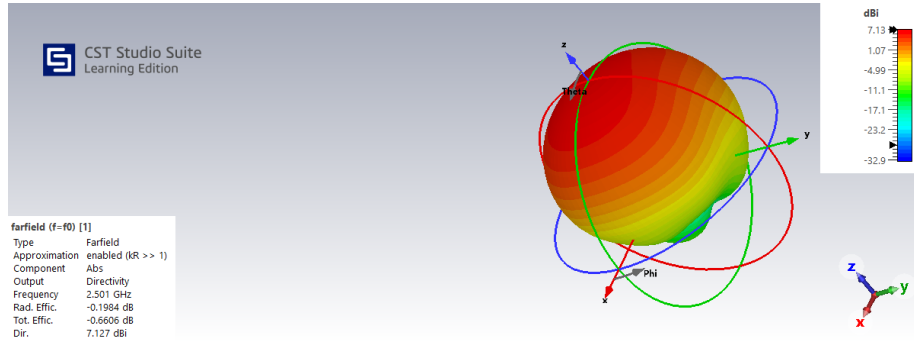


Figure 1.17: Array farfield response for excitation of Port 1 only. This result is used as one of the two element responses in the combined beam-steering post-processing.

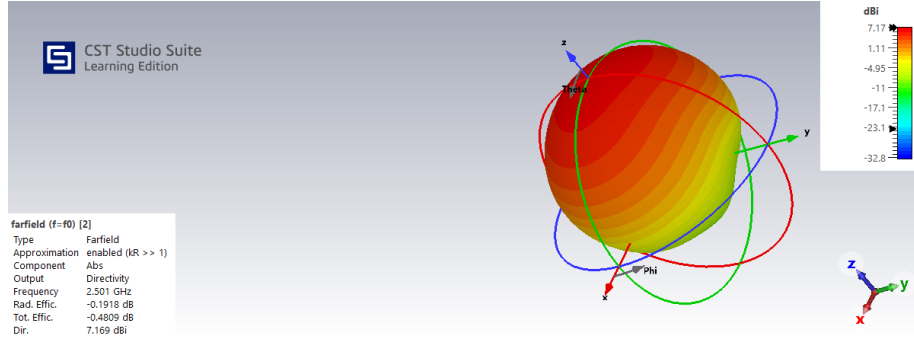


Figure 1.18: Array farfield response for excitation of Port 2 only. The two individual excitation results are later combined using amplitude and phase weights.

Table 1.16: Two-element array final simulation summary.

Metric	Final CST Value
Target frequency	2.501 GHz
Final array <code>patchL</code>	36.70 mm
Final array <code>insetY</code>	13.35 mm
Element spacing, d	59.935 mm
S_{11} near target	≈ -11.93 dB near 2.498 GHz
S_{22} near target	≈ -15.20 dB near 2.497 GHz
Mutual coupling, S_{12}	≈ -15.19 dB near 2.494 GHz
Mutual coupling, S_{21}	≈ -14.57 dB near 2.499 GHz
Combined broadside peak directivity	≈ 9.655 dBi (CST displayed value)
Metal model for tuned run	PEC approximation
Dielectric loss	Retained through $\tan \delta = 0.0009$

1.11 Beam Steering Theory and CST Implementation

Beam steering is achieved by applying a relative phase difference between the two patch elements. For a uniform two-element array with spacing d , the approximate array-factor steering relation is

$$\Delta\phi = -k_0 d \sin \theta_0.$$

For the selected spacing $d \approx \lambda_0/2$, this becomes

$$\Delta\phi \approx -180^\circ \sin \theta_0.$$

Therefore, a -90° phase shift corresponds to an ideal scan-angle magnitude of approximately 30° , while reversing the phase shift should steer the beam into the opposite half-plane. The sign displayed in CST depends on the port numbering and the selected $\phi = 0^\circ/180^\circ$ principal-plane convention.

The Python verification in Appendix B evaluates the phase-shift relation and prints the theoretical scan angles for the three CST excitation cases. The console output in Listing B.2 provides the numerical audit trail for the beam-steering cases used in the CST post-processing.

1.11.1 CST Combined-Excitation Method

The two-port array was solved in CST with Port 1 and Port 2 as separate excitations. CST then produced individual farfield responses for each port. To demonstrate beam-forming, these farfields were combined using the CST template-based post-processing option *Combine Results using am-ph Parameters* [3]. This created combined farfield results of the form

$$\text{Port 1} = 1\angle 0^\circ, \quad \text{Port 2} = 1\angle \Delta\phi.$$

The three combined cases documented in this report are listed in Table 1.17.

Table 1.17: Beam-steering cases implemented with CST amplitude-phase post-processing.

Case	CST excitation	Expected behavior
Broadside	$P_1 = 1\angle 0^\circ, \quad P_2 = 1\angle 0^\circ$	The main beam should remain near broadside.
Negative phase test	$P_1 = 1\angle 0^\circ, \quad P_2 = 1\angle -90^\circ$	The main beam should steer away from broadside into one half-plane; the ideal array-factor angle magnitude is approximately 30° .
Positive phase test	$P_1 = 1\angle 0^\circ, \quad P_2 = 1\angle +90^\circ$	The main beam should steer into the opposite half-plane with approximately the same ideal angle magnitude.

CST issued a warning that unnormalized balance was unavailable for the combined farfield results. The warning identified the combined-result radiation efficiency and gain as inaccurate. Therefore, those two quantities are not used as validated numerical results. The displayed directivity and the principal-plane pattern cuts are used to document beam shape, beam direction, and relative steering. The direct two-port S -parameters and single-patch efficiencies remain the quantitative reference results reported in Sections 1.7 and 1.10.

1.11.2 Combined Three-Dimensional Farfield Results

Figure 1.19 shows the equal-amplitude, in-phase combined farfield result. This is the reference broadside excitation. Figures 1.20 and 1.21 show the two phase-shifted cases with the CST numerical information boxes retained. The movement of the main radiation lobe confirms that the 2×1 antenna operates as a phased array when the relative port phase is changed.

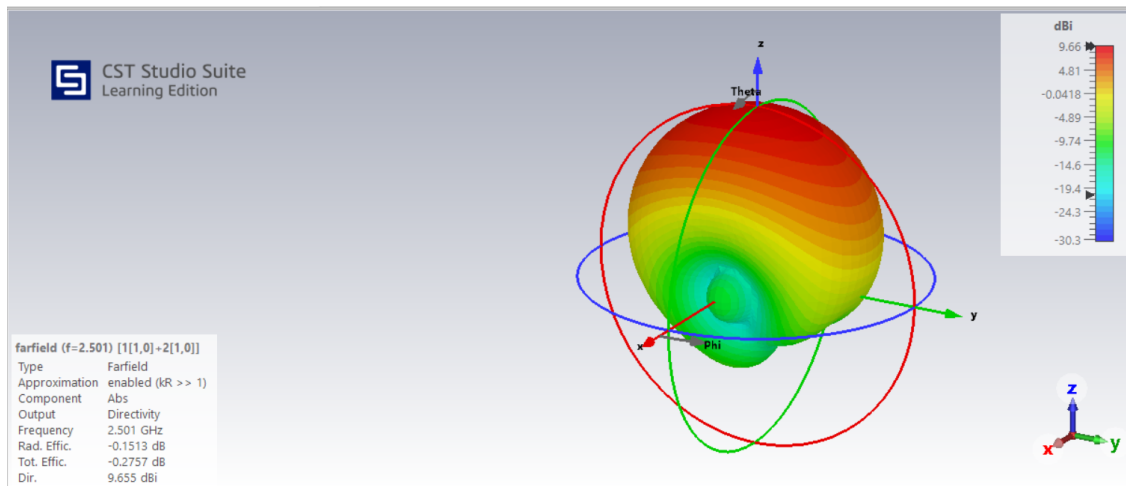


Figure 1.19: Combined 3D farfield of the tuned 2×1 array for equal-amplitude, in-phase excitation ($1\angle 0^\circ$, $1\angle 0^\circ$). CST displayed a peak directivity of approximately 9.655 dBi. The combined-result efficiency values shown in the information box are not used quantitatively because CST reported that unnormalized balance was unavailable.

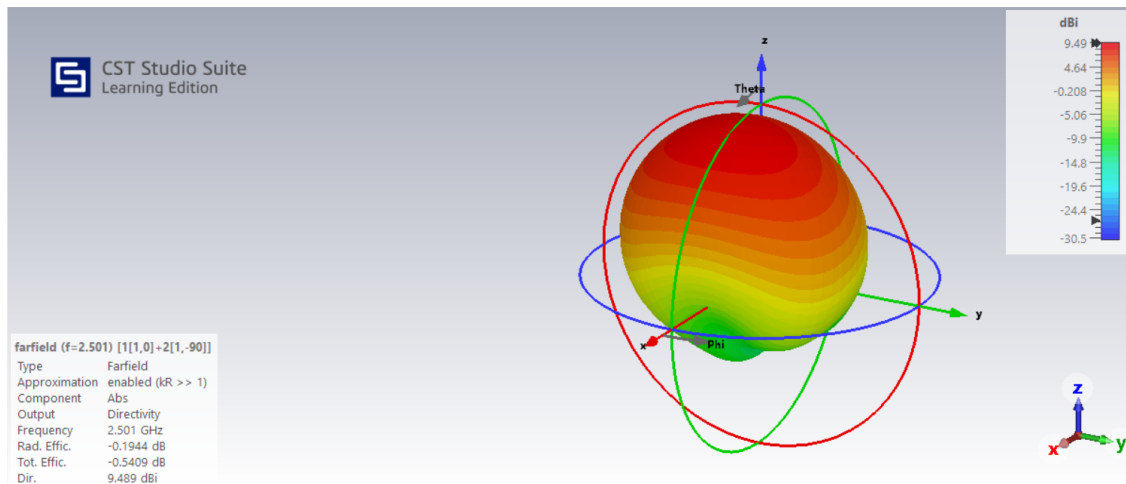


Figure 1.20: Combined 3D farfield for ($1\angle 0^\circ$, $1\angle -90^\circ$). CST displayed a peak directivity of approximately 9.489 dBi, and the main lobe moved away from the broadside reference.

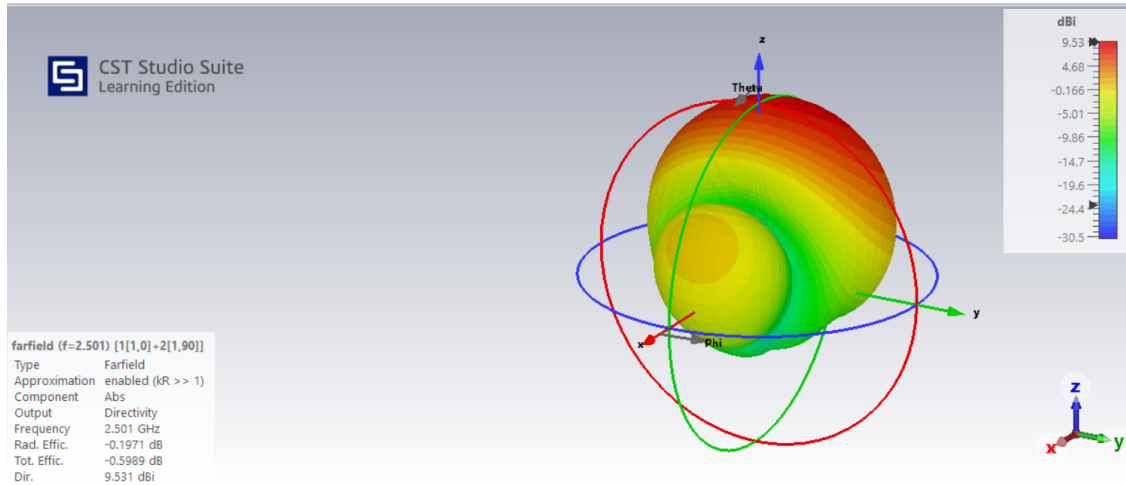


Figure 1.21: Combined 3D farfield for $(1\angle 0^\circ, 1\angle +90^\circ)$. CST displayed a peak directivity of approximately 9.531 dBi, and reversing the phase shift moved the main lobe into the opposite half-plane.

1.11.3 Combined Two-Dimensional Principal-Plane Patterns

The combined broadside pattern was also evaluated in the two principal planes. The $\phi = 0^\circ$ cut in Figure 1.22 has a main-lobe direction of 0.0° , a peak directivity of approximately 9.66 dBi, a 3-dB beamwidth of 47.0° , and a side-lobe level of approximately -18.8 dB. The $\phi = 90^\circ$ cut in Figure 1.23 has a main-lobe direction of 1.0° , a 3-dB beamwidth of 80.4° , and a side-lobe level of approximately -17.8 dB. These two cuts verify the broadside reference and quantify the unequal beamwidths in the two principal planes.

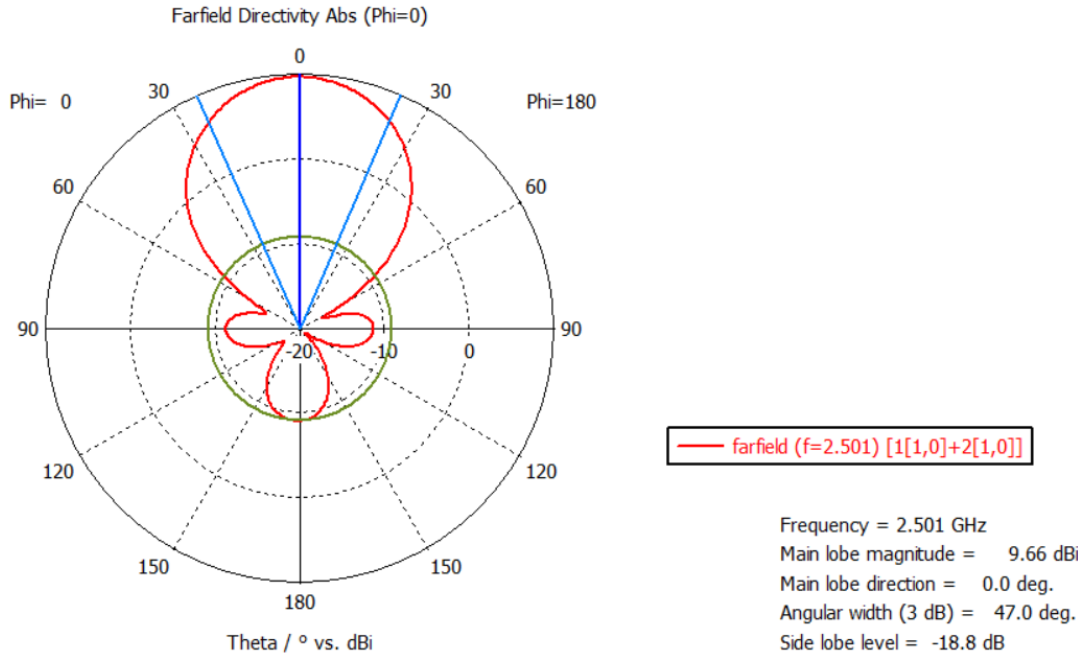


Figure 1.22: Combined broadside directivity pattern in the $\phi = 0^\circ$ plane. The main beam is at 0.0° , with peak directivity 9.66 dBi, 3-dB beamwidth 47.0° , and side-lobe level -18.8 dB.

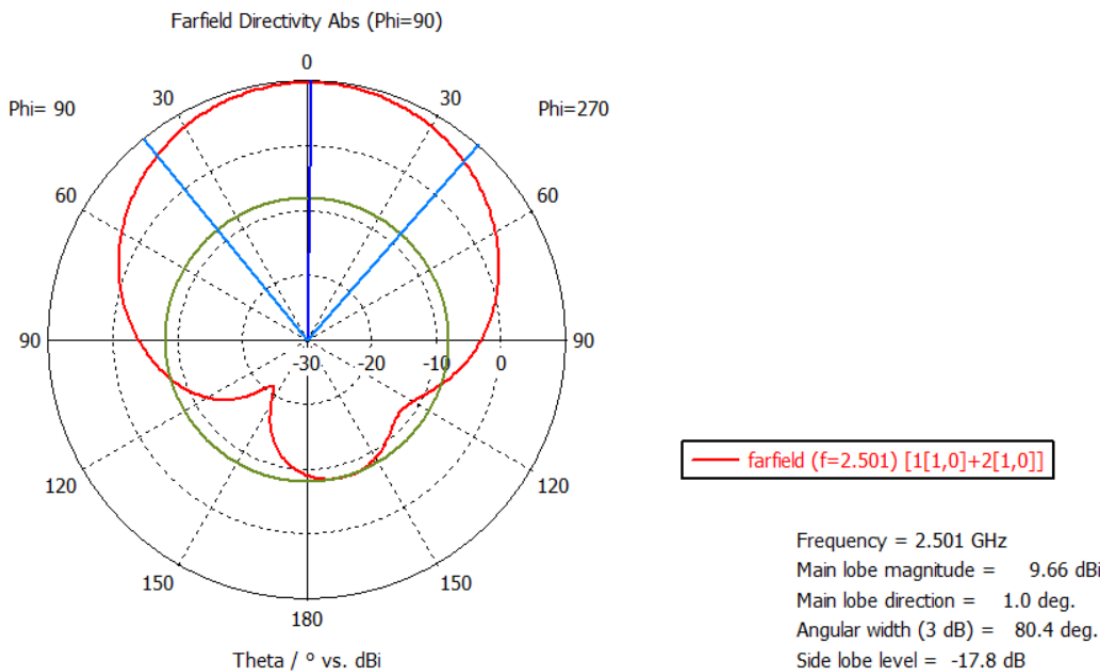


Figure 1.23: Combined broadside directivity pattern in the $\phi = 90^\circ$ plane. The main beam is at approximately 1.0° , with peak directivity 9.66 dBi, 3-dB beamwidth 80.4° , and side-lobe level -17.8 dB.

Figures 1.24 and 1.25 provide the direct quantitative beam-steering evidence. CST reports an angular magnitude of approximately 20° for both phase-shifted cases. The -90° case steers toward the $\phi = 0^\circ$ half-plane, whereas the $+90^\circ$ case steers toward the $\phi = 180^\circ$ half-plane. Thus, reversing the relative phase reverses the scan direction. The simulated 20° scan magnitude is smaller than the ideal 30° array-factor prediction because the CST result includes the finite patch element pattern, mutual coupling, substrate loading, finite ground plane, and practical feed geometry.

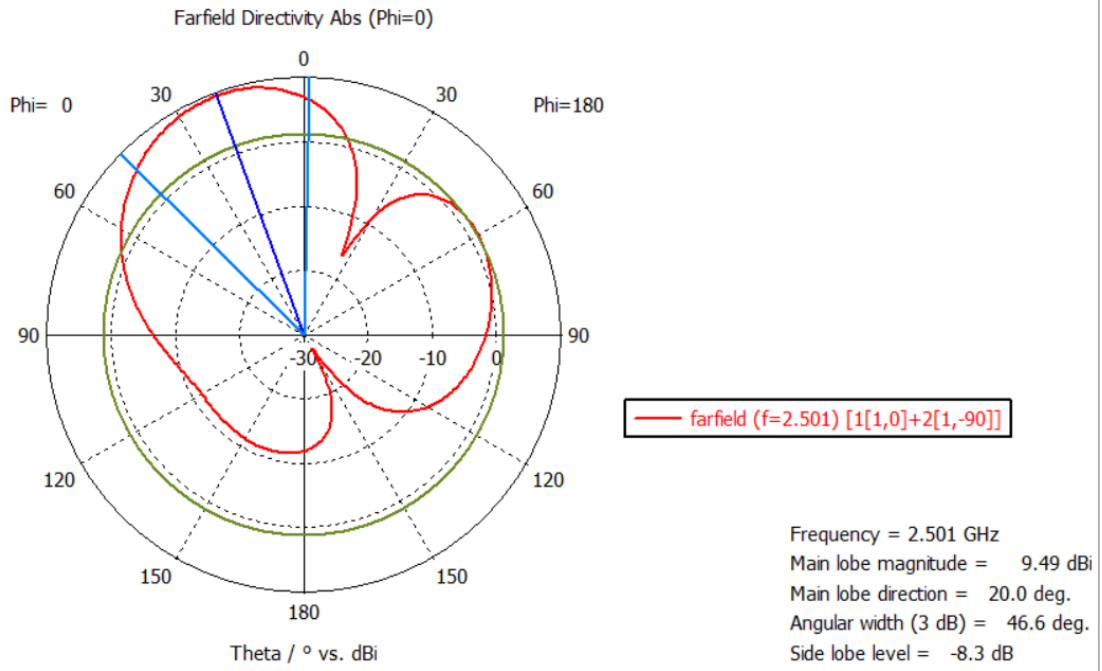


Figure 1.24: Combined $\phi = 0^\circ/180^\circ$ plane pattern for $(1\angle 0^\circ, 1\angle -90^\circ)$. The main lobe is displaced by approximately 20.0° toward the $\phi = 0^\circ$ half-plane, with displayed peak directivity 9.49 dBi and 3-dB beamwidth 46.6° .

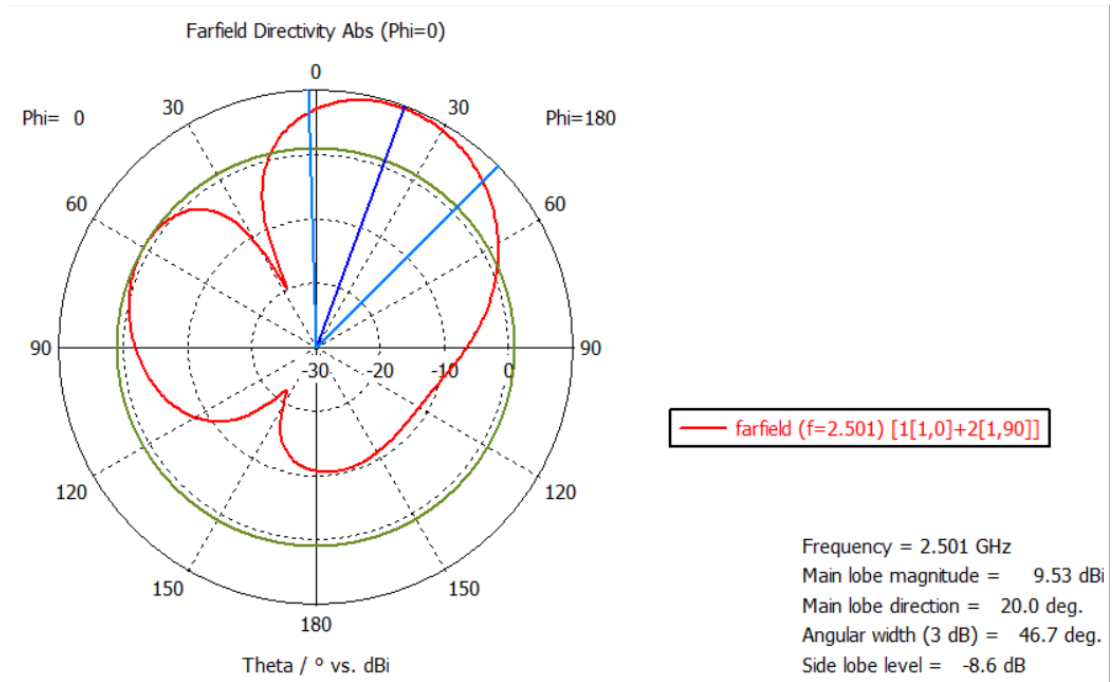


Figure 1.25: Combined $\phi = 0^\circ/180^\circ$ plane pattern for $(1\angle 0^\circ, 1\angle +90^\circ)$. The main lobe is displaced by approximately 20.0° toward the $\phi = 180^\circ$ half-plane, with displayed peak directivity 9.53 dBi and 3-dB beamwidth 46.7° .

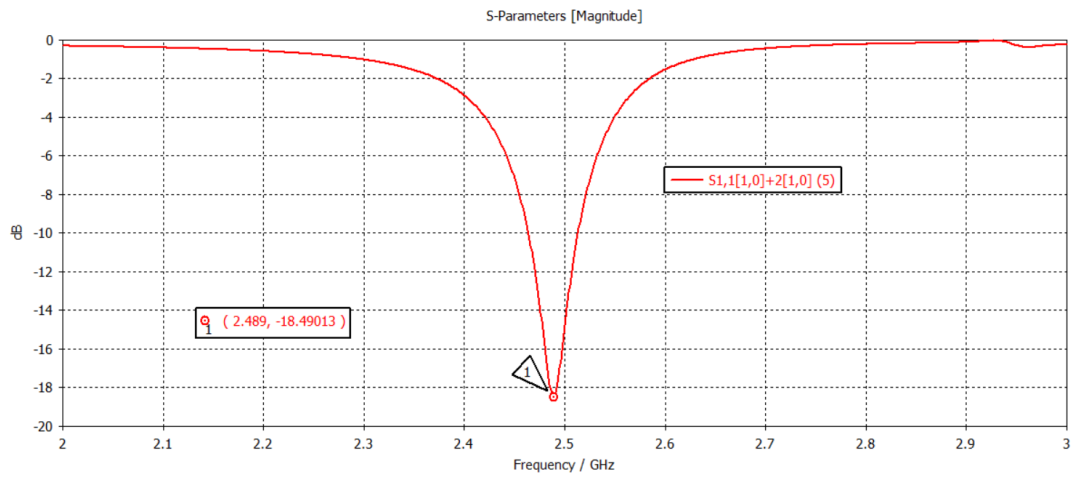


Figure 1.26: CST combined-result S -parameter output for the in-phase broadside excitation case. This post-processing result verifies that the amplitude-phase combination step was created in CST.

Table 1.18: Beam-steering result summary.

Case	Applied $\Delta\phi$	Ideal scan	CST simulated scan	CST observation
Broadside	0°	0°	0° – 1°	Broadside reference; displayed peak directivity 9.655 dBi.
Negative phase test	-90°	30° magnitude	20.0° toward $\phi = 0^\circ$	The main lobe steered into one half-plane; displayed peak directivity 9.489 dBi.
Positive phase test	$+90^\circ$	30° magnitude	20.0° toward $\phi = 180^\circ$	The beam steered into the opposite half-plane; displayed peak directivity 9.531 dBi.

1.12 Engineering Discussion

The main engineering value of this project is the connection between analytical antenna theory, full-wave CST simulation, and phased-array interpretation. The initial rectangular-patch dimensions were estimated using the transmission-line model, whereas CST Studio Suite was required to account for the finite substrate and ground-plane dimensions, practical feed geometry, fringing fields, dielectric loss, port discontinuities, and mutual coupling between the array elements.

1.12.1 Mesh-Limited CST Modeling Decision

The CST model was initially prepared using finite-conductivity copper with

$$\sigma_{\text{Cu}} = 5.87 \times 10^7 \text{ S/m.}$$

However, the finite copper thickness of 0.035 mm introduced model dimensions that were very small compared with the overall antenna size. A conventional volumetric conductor representation produced a severe edge-length ratio and exceeded the CST Learning Edition mesh-cell limit during refinement. The final baseline single-patch and array tuning runs therefore used perfect electric conductor (PEC) metallization while retaining the DiClad 880-IM dielectric loss through

$$\tan \delta = 0.0009.$$

This choice allowed the required resonance, matching, pattern, coupling, and beam-steering studies to be completed within the available computational restriction.

After the baseline design was finalized, a supplementary finite-conductivity validation was performed for the single patch using CST's *Lossy metal* model. An adaptive-mesh

attempt again exceeded the 20,000-cell limit before convergence and was discarded. A fixed-mesh run with the same geometry completed successfully and produced a resonance near 2.504 GHz, $S_{11,\min} \approx -39.72$ dB, and directivity near 7.23 dBi. These values were nearly identical to the PEC baseline. The radiation and total efficiencies decreased to approximately 92.46% and 92.38%, respectively, compared with 96.3% and 96.2% for PEC. The detailed comparison is provided in Section 1.7.4.

The validation demonstrates that PEC was an effective approximation for resonance, input matching, directivity, and normalized pattern shape in the present design. Its principal limitation was an overestimate of efficiency because finite copper ohmic loss was omitted. The fixed-mesh copper result should not be interpreted as a formal mesh-converged solution, but it provides a useful conductor-model sensitivity check within the Learning Edition limit.

1.12.2 Single-Patch Tuning Behavior

The single-patch tuning process demonstrated a physically consistent relationship between patch length and resonant frequency. The initial analytical patch length of 39.657 mm produced a resonance near 2.431 GHz, below the selected design frequency of 2.501 GHz. Because the patch was electrically too long, the patch length was reduced in successive tuning runs.

The final single-patch design used

$$L = 38.40 \text{ mm}, \quad y_{\text{inset}} = 14.11 \text{ mm},$$

and produced

$$S_{11,\min} \approx -39.87 \text{ dB}$$

at approximately

$$f_r = 2.50507 \text{ GHz}.$$

The resonant-frequency error relative to the selected design target was therefore approximately 0.16%.

The simulated single-patch farfield exhibited the expected broadside radiation behavior of a rectangular microstrip patch. The PEC baseline peak directivity was approximately 7.226 dBi, with radiation and total efficiencies of approximately 96.3% and 96.2%, respectively. The supplementary finite-conductivity run preserved essentially the same resonance, matching, directivity, and normalized pattern while reducing the radiation and total efficiencies to approximately 92.46% and 92.38%. This behavior is physically consistent with the addition of copper ohmic loss.

1.12.3 Single Patch Versus Array Tuning

When the tuned single patch was duplicated to form the 2×1 array, the resonance shifted substantially downward. The initial array configuration, which retained the final isolated-patch dimensions, produced resonance near 2.399 GHz.

This shift occurred because the individual elements no longer radiated in isolation. Mutual coupling altered the current distribution, input impedance, and effective electrical length of each patch. In addition, the enlarged common substrate and ground-plane environment modified the electromagnetic loading of the elements. Consequently, the optimized dimensions of the isolated patch could not be used directly as the final dimensions of the array.

The array was therefore retuned independently. The final array geometry used

$$L = 36.70 \text{ mm}, \quad y_{\text{inset}} = 13.35 \text{ mm}, \quad d = 59.935 \text{ mm}.$$

The selected spacing corresponds approximately to

$$d \approx \frac{\lambda_0}{2}$$

at the selected operating frequency.

Near the target frequency, the final tuned array produced approximately

$$S_{11} \approx -11.9 \text{ dB}, \quad S_{22} \approx -15.2 \text{ dB}.$$

The mutual coupling was approximately

$$S_{12} \approx S_{21} \approx -15 \text{ dB}.$$

The final array was selected as an engineering compromise among resonant-frequency proximity, acceptable matching at both input ports, and manageable mutual coupling. The coupling result confirms that the two patches are not electromagnetically independent. Excitation of one element produces fields and induced currents on the neighboring element, thereby affecting the input match, radiation behavior, and phased-array response.

1.12.4 Beam-Steering Interpretation

Beam steering was implemented using CST amplitude-phase post-processing. The independently simulated farfields associated with the two ports were combined using equal amplitudes and different relative phases.

The three evaluated excitation cases were

$$(1\angle 0^\circ, 1\angle 0^\circ),$$

$$(1\angle 0^\circ, 1\angle -90^\circ),$$

and

$$(1\angle 0^\circ, 1\angle +90^\circ).$$

The 0° relative-phase case produced the broadside reference pattern, with the two principal cuts locating the main beam at approximately 0° and 1° . Applying a relative phase shift of -90° displaced the main lobe by approximately 20° toward the $\phi = 0^\circ$ half-plane, whereas reversing the relative phase to $+90^\circ$ produced an approximately 20° displacement toward the $\phi = 180^\circ$ half-plane.

For a two-element array with approximately half-wavelength spacing, the ideal array-factor relationship is

$$\Delta\phi = -k_0 d \sin \theta_0.$$

With

$$d \approx \frac{\lambda_0}{2},$$

this becomes

$$\Delta\phi \approx -180^\circ \sin \theta_0.$$

Therefore, relative phase shifts of -90° and $+90^\circ$ ideally correspond to equal scan-angle magnitudes of approximately 30° in opposite directions. The CST-observed magnitude of approximately 20° is smaller than the ideal prediction.

Exact agreement between the ideal array-factor prediction and the CST farfield is not necessarily expected. The analytical array factor assumes idealized point sources, whereas the CST model includes the finite patch element pattern, substrate loading, feed geometry, port definition, finite ground plane, and mutual coupling. In addition, the displayed sign of the steering direction depends on the CST coordinate system and the ordering of Ports 1 and 2.

The CST post-processing results nevertheless demonstrated the essential phased-array mechanism: the direction of the combined main beam was controlled by the relative excitation phase while the excitation amplitudes remained equal.

CST reported that unnormalized balance information was unavailable for the combined post-processed farfield results. Consequently, the combined-result gain and efficiency values were not treated as validated quantitative metrics. CST displayed peak directivities of approximately 9.655 dBi, 9.489 dBi, and 9.531 dBi for the broadside, -90° , and $+90^\circ$ cases, respectively. The combined farfields and principal-plane cuts were used primarily to evaluate directivity-pattern shape, beamwidth, and the direction of beam steering.

1.12.5 Numerical Reliability and Convergence Considerations

The interpretation of the CST results must account for the numerical limitations of the available simulation environment. The PEC baseline tuning runs and the successful finite-conductivity validation used a fixed tetrahedral mesh so that the models remained within the CST Learning Edition cell limit. A formal mesh-convergence study was therefore not completed.

A separate adaptive-mesh finite-conductivity run was attempted as a numerical reliability check. The solver completed several adaptive passes, but the mesh exceeded the 20,000-cell license limit before convergence was reached. No quantitative result from that incomplete run was used in the report. The successful lossy-metal result was obtained only after mesh adaptation was disabled and is therefore identified explicitly as a fixed-mesh conductor-model sensitivity study.

A formal boundary-convergence study was also not performed. The final simulations used open boundaries with added space, but the boundary distance was not systematically increased and compared across multiple runs. Accordingly, the reported results should be interpreted as engineering-level simulation results rather than fully converged production-level predictions.

Despite these limitations, the principal physical trends were internally consistent. For the isolated single patch, reducing the patch length shifted the resonance upward toward the target frequency. For the array, adding the second element caused a downward resonance shift accompanied by quantifiable mutual coupling. The separate PEC and finite-conductivity single-patch models produced nearly identical resonance, matching, directivity, and normalized pattern shape, while the finite-conductivity model produced the expected reduction in efficiency. No unexplained non-monotonic trend was used as the basis for the final design.

A more rigorous numerical validation procedure would repeat the final single-patch and array models using several progressively refined local meshes. For each mesh level, the following quantities should be compared:

- resonant frequency;
- minimum S_{11} and S_{22} ;
- mutual coupling S_{12} and S_{21} ;
- peak directivity;
- radiation and total efficiencies; and
- beam-steering angle.

Boundary convergence could be evaluated by repeating the simulations with progressively increased open-boundary spacing and confirming that the changes in resonant frequency, S -parameters, and farfield quantities become negligible. A higher-capacity solver license would also permit adaptive finite-conductivity validation of the complete 2×1 array.

Within the computational limitations of the CST Learning Edition, the analytical calculations, single-element tuning, conductor-model sensitivity check, two-port array characterization, mutual-coupling results, and phased farfield combinations were mutually consistent. The final model therefore provides a credible demonstration of the complete engineering workflow from analytical patch design to phase-controlled array beam steering.

1.13 Engineering Evidence Matrix

Table 1.19 maps the principal technical claims in this report to the supporting analytical or simulated evidence. The matrix is intended as a traceability aid; it does not represent experimental certification. Unless explicitly identified as analytical or documentary evidence, antenna-performance results are obtained from CST full-wave simulation.

Table 1.19: Traceability matrix for the principal engineering claims and evidence.

Verification Item	Evidence in Report	Evidence Class
Design inputs and material properties	Section 1.3 and the Rogers laminate reference	Design input
Initial patch synthesis	Equations and dimensions in Sections 1.4 and 1.5	Analytical
Parameterized single-patch geometry	Figures 1.1 and 1.2; Tables 1.7 and 1.8	Simulation model
Single-patch resonance and matching	Figures 1.3 and 1.4; Table 1.10	Simulated
Single-patch directivity, patterns, and efficiencies	Figures 1.5–1.8; Table 1.10	Simulated
Finite-conductivity sensitivity	Figures 1.9–1.11; Table 1.11	Simulated sensitivity
Two-element array geometry	Figures 1.12 and 1.13; Tables 1.13 and 1.14	Simulation model
Two-port matching and mutual coupling	Figures 1.15 and 1.16; Table 1.16	Simulated

Verification Item	Evidence in Report	Evidence Class
Broadside and steered farfields	Figures 1.19–1.25; Table 1.18	Simulated post-processing
Numerical reliability and limitations	Section 1.12.5	Documented limitation
Reproducibility calculations	Appendices A and B	Analytical code

1.14 Conclusion

This project designed and simulated a 5G microstrip patch antenna and a 2×1 phased patch array. The selected design frequency was $f_0 = 2.501$ GHz, the substrate was DiClad 880-IM, and the substrate thickness was $h = 1.524$ mm. The analytical transmission-line model provided the initial patch dimensions, and CST Studio Suite was then used to tune and evaluate the final geometry.

The final tuned single patch used `patchL` = 38.40 mm and `insetY` = 14.11 mm. The PEC baseline resonance occurred at approximately 2.50507 GHz, with $S_{11,\min} \approx -39.87$ dB. The peak directivity was approximately 7.226 dBi, and the radiation and total efficiencies were approximately 96.3% and 96.2%, respectively. A supplementary fixed-mesh finite-conductivity copper run produced $f_r \approx 2.504$ GHz, $S_{11,\min} \approx -39.72$ dB, and directivity near 7.23 dBi, while the radiation and total efficiencies decreased to approximately 92.46% and 92.38%. This comparison confirmed that the PEC approximation preserved the principal resonance and radiation behavior but overestimated efficiency by omitting finite-conductor loss.

The design was then extended to a 2×1 array with an element spacing of $d = 59.935$ mm. Because mutual coupling shifted the resonance relative to the isolated single patch, the array was retuned to `patchL` = 36.70 mm and `insetY` = 13.35 mm. The final array achieved acceptable two-port matching near the operating frequency, with $S_{11} \approx -11.93$ dB, $S_{22} \approx -15.20$ dB, and mutual coupling of approximately $S_{12} \approx S_{21} \approx -15$ dB. The coupling result explains why the array required independent tuning after the isolated element was optimized.

Beam steering was demonstrated using CST amplitude-phase post-processing. The two port farfields were combined for equal-amplitude excitations with relative phases of 0° , -90° , and $+90^\circ$. The 0° case produced the broadside reference pattern, with the principal-plane main-beam direction remaining within approximately 0° – 1° . The -90° and $+90^\circ$ cases produced simulated beam displacements of approximately 20° toward the $\phi = 0^\circ$ and $\phi = 180^\circ$ half-planes, respectively. For an element spacing

of approximately $\lambda_0/2$, the ideal array-factor model predicts a scan-angle magnitude near 30° . The difference is consistent with the finite element pattern, mutual coupling, substrate loading, finite ground plane, and practical feed geometry included in the full-wave model.

The numerical results must be interpreted within the limitations of the CST Learning Edition environment. The principal tuning and array simulations used PEC metallization, while the supplementary single-patch validation used finite-conductivity lossy-metal copper. An attempted adaptive-mesh copper run exceeded the 20,000-cell limit before convergence and was discarded; the successful copper comparison therefore used a fixed mesh. Formal mesh-convergence and boundary-convergence studies were not completed. Nevertheless, the observed trends were physically consistent: reducing the isolated-patch length shifted the resonance upward, forming the array produced a downward resonance shift accompanied by quantifiable mutual coupling, and including copper loss reduced efficiency without materially changing resonance, matching, directivity, or normalized pattern shape.

A higher-fidelity validation study should repeat the final single-patch and array simulations using progressively refined local meshes and increased open-boundary spacing. The resulting resonant frequency, S -parameters, directivity, efficiency, and beam-steering angle should then be compared to confirm numerical convergence. A higher-capacity solver license would also permit adaptive finite-conductivity validation of the complete two-element array.

The public project package connects the selected design inputs, analytical patch synthesis, CST tuning, mutual-coupling analysis, numerical-modeling limitations, and phase-controlled beam steering into a traceable antenna-engineering workflow. Selected simulation figures, analytical scripts, parameter tables, and reproducibility notes accompany the report. Native CST project files are not included in the ordinary source archive because of file-size and distribution considerations.

Appendix A

Python Patch Design Verification

This appendix contains the Python script used to verify the initial rectangular microstrip patch design calculations. The purpose of this appendix is not only to include source code, but also to document the exact console output used to audit the numerical values reported in the theory and analytical-design sections.

```

1  """
2  Initial rectangular microstrip patch design calculations.
3
4  This script verifies the analytical starting dimensions used in the
5  public engineering report. The selected design frequency is 2.501 GHz.
6  The substrate is DiClad 880-IM with h = 0.060 in = 1.524 mm.
7  """
8
9  from __future__ import annotations
10
11  from math import acos, log, pi, sqrt
12
13  C0 = 299_792_458.0  # Speed of light in vacuum, m/s
14  F0_HZ = 2.501e9
15
16  # Nominal Rogers DiClad 880 design values.
17  EPS_R = 2.20
18  TAN_DELTA = 0.0009
19  H_M = 1.524e-3
20
21  # Inset-feed engineering starting assumptions.
22  Z0_TARGET = 50.0
23  EDGE_RESISTANCE_ESTIMATE = 300.0
24  INSET_GAP_M = 1.000e-3
25  FEED_EXTENSION_M = 15.000e-3
26
27
28  def microstrip_effective_eps(width_over_h: float, eps_r: float) -> float:
29      """Return the quasi-static effective permittivity of a microstrip line."""

```

```

30     return (
31         (eps_r + 1.0) / 2.0
32         + (eps_r - 1.0) / 2.0
33         * (1.0 + 12.0 / width_over_h) ** (-0.5)
34     )
35
36
37 def microstrip_z0_from_u(width_over_h: float, eps_r: float) -> float:
38     """Return the characteristic impedance for a given width ratio W/h."""
39     u = width_over_h
40     eps_eff = microstrip_effective_eps(u, eps_r)
41
42     if u <= 1.0:
43         return 60.0 / sqrt(eps_eff) * log(8.0 / u + 0.25 * u)
44
45     return 120.0 * pi / (
46         sqrt(eps_eff) * (u + 1.393 + 0.667 * log(u + 1.444))
47     )
48
49
50 def solve_microstrip_width_for_z0(
51     z0_target: float,
52     eps_r: float,
53     h_m: float,
54 ) -> tuple[float, float, float]:
55     """Solve for microstrip width using bisection on the ratio W/h."""
56     lo = 0.01
57     hi = 20.0
58
59     for _ in range(100):
60         mid = 0.5 * (lo + hi)
61         z_mid = microstrip_z0_from_u(mid, eps_r)
62
63         if z_mid > z0_target:
64             lo = mid
65         else:
66             hi = mid
67
68     u = 0.5 * (lo + hi)
69     width_m = u * h_m
70     eps_eff_feed = microstrip_effective_eps(u, eps_r)
71     return width_m, u, eps_eff_feed
72
73
74 lambda0_m = C0 / F0_HZ
75 patch_width_m = C0 / (2.0 * F0_HZ) * sqrt(2.0 / (EPS_R + 1.0))
76 eps_eff = (
77     (EPS_R + 1.0) / 2.0
78     + (EPS_R - 1.0) / 2.0
79     * (1.0 + 12.0 * H_M / patch_width_m) ** (-0.5)
80 )
81 delta_l_m = (

```



```

82     0.412
83     * H_M
84     * ((eps_eff + 0.3) * (patch_width_m / H_M + 0.264))
85     / ((eps_eff - 0.258) * (patch_width_m / H_M + 0.8))
86 )
87 effective_length_m = C0 / (2.0 * F0_HZ * sqrt(eps_eff))
88 patch_length_m = effective_length_m - 2.0 * delta_l_m
89
90 # Finite ground-plane and substrate starting dimensions.
91 ground_width_m = patch_width_m + 6.0 * H_M
92 minimum_ground_length_m = patch_length_m + 6.0 * H_M
93 ground_length_with_feed_m = patch_length_m + FEED_EXTENSION_M + 6.0 * H_M
94 element_spacing_m = lambda0_m / 2.0
95
96 feed_width_m, feed_width_over_h, eps_eff_feed = (
97     solve_microstrip_width_for_z0(Z0_TARGET, EPS_R, H_M)
98 )
99 feed_z0_check = microstrip_z0_from_u(feed_width_over_h, EPS_R)
100 inset_depth_m = (
101     patch_length_m
102     / pi
103     * acos(sqrt(Z0_TARGET / EDGE_RESISTANCE_ESTIMATE))
104 )
105
106 print("Initial Microstrip Patch Design Verification")
107 print("=" * 72)
108 print(f"Selected f0: {F0_HZ / 1e9:.6f} GHz")
109 print("Selected substrate: DiClad 880-IM")
110 print(f"Working eps_r: {EPS_R:.4f}")
111 print(f"Working tan_delta: {TAN_DELTA:.5f}")
112 print(f"Substrate thickness h: {H_M * 1e3:.3f} mm")
113 print("-" * 72)
114 print("Rectangular patch starting dimensions")
115 print(f"lambda0: {lambda0_m * 1e3:.3f} mm")
116 print(f"Patch width W: {patch_width_m * 1e3:.3f} mm")
117 print(f"Effective permittivity eps_eff: {eps_eff:.4f}")
118 print(f"Length extension DeltaL: {delta_l_m * 1e3:.3f} mm")
119 print(f"Effective length L_eff: {effective_length_m * 1e3:.3f} mm")
120 print(f"Patch length L: {patch_length_m * 1e3:.3f} mm")
121 print("-" * 72)
122 print("Finite ground/substrate starting dimensions")
123 print(f"Minimum ground width Wg = W + 6h: {ground_width_m * 1e3:.3f} mm")
124 print(
125     "Minimum ground length Lg_min = L + 6h: "
126     f"{minimum_ground_length_m * 1e3:.3f} mm"
127 )
128 print(f"Feed extension Lf: {FEED_EXTENSION_M * 1e3:.3f} mm")
129 print(
130     "Inset-fed model ground length Lg = L + Lf + 6h: "
131     f"{ground_length_with_feed_m * 1e3:.3f} mm"
132 )
133 print("-" * 72)

```

```

134 print("Inset-fed microstrip matching starting values")
135 print(f"Target feed-line impedance Z0: {Z0_TARGET:.1f} ohm")
136 print(f"Feed width ratio Wf/h: {feed_width_over_h:.4f}")
137 print(f"Feed-line effective permittivity eps_eff_feed: {eps_eff_feed:.4f}")
138 print(f"Feed-line width Wf: {feed_width_m * 1e3:.3f} mm")
139 print(f"Feed-line Z0 verification: {feed_z0_check:.3f} ohm")
140 print(f"Assumed patch edge resistance: {EDGE_RESISTANCE_ESTIMATE:.1f} ohm")
141 print(f"Initial inset depth y0: {inset_depth_m * 1e3:.3f} mm")
142 print(f"Initial inset gap gi: {INSET_GAP_M * 1e3:.3f} mm")
143 print("-" * 72)
144 print("Array starting value")
145 print(f"Initial array spacing d=lambda0/2: {element_spacing_m * 1e3:.3f} mm")

```

Listing A.1: Python script for verifying the initial rectangular patch design dimensions.

A.1 Console Output

```

Initial Microstrip Patch Design Verification
=====
Selected f0: 2.501000 GHz
Selected substrate: DiClad 880-IM
Working eps_r: 2.2000
Working tan_delta: 0.00090
Substrate thickness h: 1.524 mm
-----

Rectangular patch starting dimensions
lambda0: 119.869 mm
Patch width W: 47.382 mm
Effective permittivity eps_eff: 2.1097
Length extension DeltaL: 0.803 mm
Effective length L_eff: 41.264 mm
Patch length L: 39.657 mm
-----

Finite ground/substrate starting dimensions
Minimum ground width Wg = W + 6h: 56.526 mm
Minimum ground length Lg_min = L + 6h: 48.801 mm
Feed extension Lf: 15.000 mm
Inset-fed model ground length Lg = L + Lf + 6h: 63.801 mm
-----

Inset-fed microstrip matching starting values
Target feed-line impedance Z0: 50.0 ohm
Feed width ratio Wf/h: 3.1069
Feed-line effective permittivity eps_eff_feed: 1.8721
Feed-line width Wf: 4.735 mm
Feed-line Z0 verification: 50.000 ohm
Assumed patch edge resistance: 300.0 ohm
Initial inset depth y0: 14.520 mm
Initial inset gap gi: 1.000 mm
-----

Array starting value
Initial array spacing d=lambda0/2: 59.935 mm

```

Listing A.2: Console output from the patch design verification script.

Appendix B

Python Beam-Steering Verification

This appendix contains the Python script used to calculate beam-steering phase shifts for the 2×1 array. The console output is included to make the CST excitation phases traceable to the analytical array-factor equation used in the main report.

```

1  """Beam-steering phase-shift calculations for a two-element patch array."""
2
3  from math import sin, asin, radians, degrees, pi
4
5  C0 = 299_792_458.0
6  f0_hz = 2.501e9
7  lambda0_m = C0 / f0_hz
8  # The CST-tuned array used arrayD = lambda0/2 from the analytical setup.
9  d_m = lambda0_m / 2.0
10 k0 = 2.0 * pi / lambda0_m
11
12 print("Two-Element Array Beam-Steering Phase Shifts")
13 print("=" * 72)
14 print(f"f0 = {f0_hz/1e9:.6f} GHz")
15 print(f"lambda0 = {lambda0_m*1e3:.3f} mm")
16 print(f"d = lambda0/2 = {d_m*1e3:.3f} mm")
17 print("-" * 72)
18 print("Desired steering angle -> required phase shift")
19 print("theta0_deg, Delta_phi_deg")
20 for theta_deg in [0, 10, 15, 20, 30, 45]:
21     delta_phi_rad = -k0 * d_m * sin(radians(theta_deg))
22     print(f"{theta_deg:10.1f}, {degrees(delta_phi_rad):14.3f}")
23
24 print("-" * 72)
25 print("Applied CST phase shift -> ideal array-factor beam angle")
26 print("Delta_phi_deg, theta0_deg")
27 for delta_phi_deg in [0, -90, +90]:
28     delta_phi_rad = radians(delta_phi_deg)
29     sin_theta = -delta_phi_rad / (k0 * d_m)
30     theta_deg = degrees(asin(sin_theta))

```

```
31 print(f"{delta_phi_deg:13.1f}, {theta_deg:10.3f}")
```

Listing B.1: Python script for calculating two-element array beam-steering phase shifts.

B.1 Console Output

```
Two-Element Array Beam-Steering Phase Shifts
=====
f0 = 2.501000 GHz
lambda0 = 119.869 mm
d = lambda0/2 = 59.935 mm
-----
Desired steering angle -> required phase shift
theta0_deg, Delta_phi_deg
    0.0,      -0.000
   10.0,     -31.257
   15.0,     -46.587
   20.0,     -61.564
   30.0,     -90.000
   45.0,    -127.279
-----
Applied CST phase shift -> ideal array-factor beam angle
Delta_phi_deg, theta0_deg
    0.0,      -0.000
   -90.0,     30.000
    90.0,    -30.000
```

Listing B.2: Console output from the beam-steering phase-shift verification script.

References

- [1] Rogers Corporation, *DiClad 870 and DiClad 880 Laminates*, Material properties used for DiClad 880: design dielectric constant 2.20 and dissipation factor 0.0009., Rogers Corporation. Accessed: Jul. 11, 2026. [Online]. Available: https://www.rogerscorp.com/advanced-electronics-solutions/diclad-series-laminates/diclad-870_880-laminates.
- [2] W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2013.
- [3] Dassault Systèmes SIMULIA, *CST Studio Suite Documentation*, Dassault Systèmes, 2026.